# Electrodynamics

# 电动力学

Version 2026.8.6

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# 目录

# 1 Maxwell 电磁场方程

## 1.1 Maxwell 方程组的导出

### 1.1.1 电荷守恒定律

**Law. 1 电荷守恒定律**

$$\boldsymbol{\nabla}\cdot\boldsymbol{J}(t,\boldsymbol{x})+\frac{\partial\rho(t,\boldsymbol{x})}{\partial t}=0 \tag{1.1}$$

**Cor. 1 稳恒电流的散度**

对于稳恒体系，物理量不显含时间，$\dfrac{\partial\rho}{\partial t}=0$, 从而

$$\boldsymbol{\nabla}\cdot\boldsymbol{J}=0 \tag{1.2}$$

**Prop. 1 稳恒电流的边值关系**

$$\boldsymbol{n}\cdot(\boldsymbol{J}_2-\boldsymbol{J}_1)=0 \quad\Longleftrightarrow\quad J_{2n}=J_{1n} \tag{1.3}$$

**Thm. 1 守恒律普遍表达式**

$$[\text{流密度的散度}]+[\text{空间密度的时间变化率}]=0$$

称为连续性方程或流守恒方程.

### 1.1.2 静电场

**Law. 2 Coulomb 定律**

$$\boldsymbol{F}=\frac{1}{4\pi\varepsilon_0}\frac{qq'}{r^3}\boldsymbol{r} \tag{1.4}$$

其中 $\boldsymbol{r}=\boldsymbol{x}-\boldsymbol{x}'$, $r=|\boldsymbol{x}-\boldsymbol{x}'|$.

**Def. 1 电场强度**

$$\boldsymbol{E}(\boldsymbol{x})\xlongequal{\text{def}}\frac{\boldsymbol{F}(\boldsymbol{x})}{q}=\frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\boldsymbol{x}')\boldsymbol{r}}{r^3}\,\mathrm{d}^3x' \tag{1.5}$$

**Lem. 1**

$$\boldsymbol{\nabla}\frac{1}{r}=-\frac{\boldsymbol{r}}{r^3} \tag{1.6}$$

$$\nabla^2\frac{1}{r}=-\boldsymbol{\nabla}\cdot\frac{\boldsymbol{r}}{r^3}=-4\pi\delta(\boldsymbol{r}) \tag{1.7}$$

$$\boldsymbol{\nabla}\times\boldsymbol{\nabla}\phi=0 \tag{1.8}$$

**Thm. 2 静电场的散度：电场的 Gauss 定理**

$$\boldsymbol{\nabla}\cdot\boldsymbol{E}(\boldsymbol{x})=\frac{\rho(\boldsymbol{x})}{\varepsilon_0} \tag{1.9}$$

**Pf** $\boldsymbol{\nabla}\cdot\boldsymbol{E}(\boldsymbol{x})=\boldsymbol{\nabla}\cdot\left(\frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\boldsymbol{x}')\boldsymbol{r}}{r^3}\,\mathrm{d}^3x'\right)=\frac{1}{4\pi\varepsilon_0}\int\rho(\boldsymbol{x}')\,\mathrm{d}^3x'\,\underbrace{\boldsymbol{\nabla}\cdot\left(\frac{\boldsymbol{r}}{r^3}\right)}_{=4\pi\delta(\boldsymbol{r})},$

$\Rightarrow\boldsymbol{\nabla}\cdot\boldsymbol{E}(\boldsymbol{x})=\frac{1}{\varepsilon_0}\int\rho(\boldsymbol{x}')\delta(\boldsymbol{r})\,\mathrm{d}^3x'=\frac{\rho(\boldsymbol{x})}{\varepsilon_0}.$

**Cor. 2 时变电场的散度**

由于未使用恒定条件，可推广为

$$\boldsymbol{\nabla}\cdot\boldsymbol{E}(t,\boldsymbol{x})=\frac{\rho(t,\boldsymbol{x})}{\varepsilon_0} \tag{1.10}$$

**Thm. 3 静电场的旋度**

$$\boldsymbol{\nabla}\times\boldsymbol{E}(\boldsymbol{x})=0 \tag{1.11}$$

**Pf** $\boldsymbol{E}(\boldsymbol{x})=\frac{1}{4\pi\varepsilon_0}\int\rho(\boldsymbol{x}')\,\mathrm{d}^3x'\,\frac{\boldsymbol{r}}{r^3}=-\frac{1}{4\pi\varepsilon_0}\int\rho(\boldsymbol{x}')\,\mathrm{d}^3x'\,\boldsymbol{\nabla}\frac{1}{r}=-\boldsymbol{\nabla}\underbrace{\left(\frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\boldsymbol{x}')}{r}\,\mathrm{d}^3x'\right)}_{\xlongequal{\text{def}}\phi(\boldsymbol{x})},$

$\Rightarrow\boldsymbol{\nabla}\times\boldsymbol{E}(\boldsymbol{x})=-\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\phi)=0.$

**Def. 2 电标势**

$$\phi(\boldsymbol{x})=\frac{1}{4\pi\varepsilon_0}\int\frac{\rho(\boldsymbol{x}')}{r}\,\mathrm{d}^3x' \tag{1.12}$$

### 1.1.3 静磁场

**Law. 3 Biot-Savart 定律**

$$\boldsymbol{B}(\boldsymbol{x})=\frac{\mu_0}{4\pi}\int\frac{\boldsymbol{J}(\boldsymbol{x}')\times\boldsymbol{r}}{r^3}\,\mathrm{d}^3x'=\frac{\mu_0}{4\pi}\oint\frac{I\,\mathrm{d}\boldsymbol{x}\times\boldsymbol{r}}{r^3} \tag{1.13}$$

**Lem. 2**

$$\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})=0 \tag{1.14}$$

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{A})=\boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{A})-\nabla^2\boldsymbol{A} \tag{1.15}$$

**Thm. 4 静磁场的散度**

$$\boldsymbol{\nabla}\cdot\boldsymbol{B}(\boldsymbol{x})=0 \tag{1.16}$$

**Pf** $\dfrac{\boldsymbol{J}\times\boldsymbol{r}}{r^3}=-\boldsymbol{J}\times\boldsymbol{\nabla}\dfrac{1}{r}=\boldsymbol{\nabla}\times\dfrac{\boldsymbol{J}}{r}-\underbrace{\dfrac{\boldsymbol{\nabla}\times\boldsymbol{J}(\boldsymbol{x}')}{r}}_{=0}$,

$$\Rightarrow \boldsymbol{B}(\boldsymbol{x})=\frac{\mu_0}{4\pi}\int\boldsymbol{\nabla}\times\frac{\boldsymbol{J}(\boldsymbol{x}')}{r}\,\mathrm{d}^3x'=\boldsymbol{\nabla}\times\underbrace{\left(\frac{\mu_0}{4\pi}\int\frac{\boldsymbol{J}(\boldsymbol{x}')}{r}\,\mathrm{d}^3x'\right)}_{\overset{\text{def}}{=\!=}\boldsymbol{A}},\Rightarrow \boldsymbol{\nabla}\cdot\boldsymbol{B}(\boldsymbol{x})=\boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{A})=0.$$

**Cor. 3 非恒定情况**

$$\boldsymbol{\nabla}\cdot\boldsymbol{B}(t,\boldsymbol{x})=0 \tag{1.17}$$

**Def. 3 磁矢势**

$$\boldsymbol{A}(\boldsymbol{x})=\frac{\mu_0}{4\pi}\int\frac{\boldsymbol{J}(\boldsymbol{x}')}{r}\,\mathrm{d}^3x' \tag{1.18}$$

**Thm. 5 静磁场的旋度：Ampère 环路定理**

$$\boldsymbol{\nabla}\times\boldsymbol{B}(\boldsymbol{x})=\mu_0\boldsymbol{J}(\boldsymbol{x}) \tag{1.19}$$

**Pf** $\boldsymbol{\nabla}\cdot\boldsymbol{A}(\boldsymbol{x})=\dfrac{\mu_0}{4\pi}\displaystyle\int\boldsymbol{\nabla}\cdot\dfrac{\boldsymbol{J}(\boldsymbol{x}')}{r}\,\mathrm{d}^3x'$. $\boldsymbol{\nabla}\cdot\dfrac{\boldsymbol{J}(\boldsymbol{x}')}{r}=\boldsymbol{\nabla}\dfrac{1}{r}\cdot\boldsymbol{J}=-\boldsymbol{\nabla}'\dfrac{1}{r}\cdot\boldsymbol{J}=\dfrac{\boldsymbol{\nabla}'\cdot\boldsymbol{J}}{r}-\boldsymbol{\nabla}'\cdot\dfrac{\boldsymbol{J}}{r}$.

由电流恒定条件，有 $\boldsymbol{\nabla}'\cdot\boldsymbol{J}(\boldsymbol{x}')=0$, $\displaystyle\int\boldsymbol{\nabla}'\cdot\frac{\boldsymbol{J}}{r}\,\mathrm{d}^3x'=\oint\frac{\boldsymbol{J}}{r}\cdot\boldsymbol{n}\,\mathrm{d}^2x'=0$, $\Rightarrow \underbrace{\boldsymbol{\nabla}\cdot\boldsymbol{A}(\boldsymbol{x})=0}_{\text{Coulomb 规范}}$.

$$\nabla^2\boldsymbol{A}(\boldsymbol{x})=\frac{\mu_0}{4\pi}\int\nabla^2\frac{1}{r}\boldsymbol{J}(\boldsymbol{x}')\,\mathrm{d}^3x'=-\mu_0\int\delta(\boldsymbol{r})\boldsymbol{J}(\boldsymbol{x}')\,\mathrm{d}^3x'=\mu_0\boldsymbol{J}(\boldsymbol{x}).\ \Rightarrow \boldsymbol{\nabla}\times\boldsymbol{B}=\mu_0\boldsymbol{J}(\boldsymbol{x}).$$

**Lem. 3 柱坐标中的旋度**

$$\boldsymbol{\nabla}\times\boldsymbol{B}=\frac{1}{r}\begin{vmatrix}\boldsymbol{e}_r & r\boldsymbol{e}_\theta & \boldsymbol{e}_z\\ \partial_r & \partial_\theta & \partial_z\\ B_r & rB_\theta & B_z\end{vmatrix} \tag{1.20}$$

### 1.1.4 感生电场

**Law. 4 Faraday 电磁感应定律**

$$\mathscr{E} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} \tag{1.21}$$

**Thm. 6 感生电场的旋度**

$$\boldsymbol{\nabla} \times \boldsymbol{E}(t, \boldsymbol{x}) = -\frac{\partial \boldsymbol{B}(t, \boldsymbol{x})}{\partial t} \tag{1.22}$$

**Rmk** 变化的磁场激发感生涡旋电场.

**Pf** $\mathscr{E} = \oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{x} = -\frac{\mathrm{d}\Phi_B}{\mathrm{d}t} = \int (\boldsymbol{\nabla} \times \boldsymbol{E}) \cdot \mathrm{d}\boldsymbol{S} = -\frac{\mathrm{d}}{\mathrm{d}t}\int \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{S} = -\int \frac{\partial \boldsymbol{B}}{\partial t} \cdot \mathrm{d}\boldsymbol{S}.$

### 1.1.5 位移电流

**Def. 4 位移电流**

$$\boldsymbol{J}_{\mathrm{D}} = \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \tag{1.23}$$

**Rmk** 变化的电场激发磁场.

**Pf** $0 = \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B}) = \mu_0 \left(\boldsymbol{\nabla} \cdot \boldsymbol{J} + \frac{\partial \rho}{\partial t}\right) = \mu_0 \left(\boldsymbol{\nabla} \cdot \boldsymbol{J} + \varepsilon_0 \frac{\partial}{\partial t} \boldsymbol{\nabla} \cdot \boldsymbol{E}\right) = \mu_0 \boldsymbol{\nabla} \cdot (\boldsymbol{J} + \boldsymbol{J}_{\mathrm{D}}).$

**Thm. 7 磁场的旋度：Maxwell-Ampère 定理**

$$\boldsymbol{\nabla} \times \boldsymbol{B}(t, \boldsymbol{x}) = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \tag{1.24}$$

### 1.1.6 真空中 Maxwell 方程组

**Thm. 8 真空中 Maxwell 方程组**

$$\begin{cases} \boldsymbol{\nabla} \cdot \boldsymbol{E} = \dfrac{\rho}{\varepsilon_0} \\ \boldsymbol{\nabla} \times \boldsymbol{E} = -\dfrac{\partial \boldsymbol{B}}{\partial t} \\ \boldsymbol{\nabla} \cdot \boldsymbol{B} = 0 \\ \boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \mu_0 \varepsilon_0 \dfrac{\partial \boldsymbol{E}}{\partial t} \end{cases} \tag{1.25}$$

## 1.2 介质中的电磁场方程

**介质对外场的响应**

外场的作用使介质分子的电偶极矩与分子电流的磁偶极矩取向呈现规律性, 导致出现宏观的电荷与电流分布, 其激发的电磁场与外场叠加即为总场.

$$\begin{cases} \rho_{\text{total}} = \rho + \rho_{\text{P}} \\ \boldsymbol{J}_{\text{total}} = \boldsymbol{J} + \boldsymbol{J}_{\text{P}} + \boldsymbol{J}_{\text{M}} \end{cases} \tag{1.26}$$

其中 $\rho, \boldsymbol{J}$ 分别为自由电荷密度与自由电流密度. 自由电荷面密度则为 $\sigma$, 线密度为 $\lambda$.

### 1.2.1 介质的极化

**Def. 5 极化强度**

$$\boldsymbol{P} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \sum_i \boldsymbol{p}_i = n\boldsymbol{p} \tag{1.27}$$

**Prop. 2 极化电荷的体分布**

$$\rho_{\text{P}} = -\boldsymbol{\nabla} \cdot \boldsymbol{P} \tag{1.28}$$

**Rmk** 均匀极化的介质内, 极化电荷只分布于自由电荷附近与介质界面处.

**Pf** $\displaystyle \int \rho_{\text{P}}(\boldsymbol{x})\, \mathrm{d}^3x = -\oint nq\boldsymbol{l} \cdot \mathrm{d}^2\boldsymbol{x} = -\oint \underbrace{\boldsymbol{P} \cdot \boldsymbol{n}}_{\overset{\text{def}}{=\!=}\sigma_{\text{P}}} \mathrm{d}^2x = -\int \boldsymbol{\nabla} \cdot \boldsymbol{P}\, \mathrm{d}^3x.$

**Prop. 3 极化电荷的面分布**

$$\sigma_{\text{P}} = -\boldsymbol{n} \cdot (\boldsymbol{P}_2 - \boldsymbol{P}_1) = -(P_{2n} - P_{1n}) = \varepsilon_0(E_{2n} - E_{1n}) - \sigma \tag{1.29}$$

其中 $\boldsymbol{n}$ 为从介质 1 指向介质 2 的单位法矢量.

**Pf** $\displaystyle \sigma_{\text{P}}\Delta S = -\oint \boldsymbol{n} \cdot \boldsymbol{P}\, \mathrm{d}^2x = -\boldsymbol{n} \cdot (\boldsymbol{P}_2 - \boldsymbol{P}_1)\Delta S.$

**Prop. 4 极化电流**

$$\boldsymbol{J}_{\text{P}} = \frac{\partial \boldsymbol{P}}{\partial t} \tag{1.30}$$

**Pf** 由电荷守恒律, $\displaystyle \boldsymbol{\nabla} \cdot \boldsymbol{J}_{\text{P}} = -\frac{\partial \rho_{\text{P}}}{\partial t} = \boldsymbol{\nabla} \cdot \frac{\partial \boldsymbol{P}}{\partial t}.$

**Cor. 4　均匀极化体的总极矩**

若介质体均匀极化, 即极化强度为常矢量 $\boldsymbol{P}$, 其体积为 $V$, 则其总极矩为

$$\boldsymbol{p}_{\text{total}} = \boldsymbol{P}V \tag{1.31}$$

**Lem. 4　二阶张量的 Gauss 公式**

$$\int_{\Omega} \boldsymbol{\nabla}\cdot\mathcal{T}\,\mathrm{d}^3x = \oint_{\partial\Omega} \boldsymbol{n}\cdot\mathcal{T}\,\mathrm{d}^2x \tag{1.32}$$

**Prop. 5　孤立介质所受电场力**

$$\boldsymbol{F} = \int \rho\boldsymbol{E}_0 + \boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{E}_0\,\mathrm{d}^3x \tag{1.33}$$

**Pf**　$\boldsymbol{F} = \int(\rho+\rho_{\mathrm{P}})\boldsymbol{E}_0\,\mathrm{d}^3x + \oint\sigma_{\mathrm{P}}\boldsymbol{E}_0\,\mathrm{d}^2x = \int(\rho-\boldsymbol{\nabla}\cdot\boldsymbol{P})\boldsymbol{E}_0\,\mathrm{d}^3x + \oint\boldsymbol{n}\cdot\boldsymbol{P}\boldsymbol{E}\,\mathrm{d}^2x$
$= \int(\rho-\boldsymbol{\nabla}\cdot\boldsymbol{P})\boldsymbol{E}_0 + \boldsymbol{\nabla}\cdot(\boldsymbol{P}\boldsymbol{E}_0)\,\mathrm{d}^3x = \int \rho\boldsymbol{E}_0 \underbrace{-(\boldsymbol{\nabla}\cdot\boldsymbol{P})\boldsymbol{E}_0 + (\boldsymbol{\nabla}\cdot\boldsymbol{P})\boldsymbol{E}_0}_{=0} + \boldsymbol{P}\cdot\boldsymbol{\nabla}\boldsymbol{E}_0\,\mathrm{d}^3x.$

### 1.2.2　介质的磁化

**Def. 6　磁化强度**

$$\boldsymbol{M} = \lim_{\Delta V\to 0}\frac{1}{\Delta V}\sum_i \boldsymbol{M}_i = n\boldsymbol{m} \tag{1.34}$$

**Prop. 6　磁化电流的体分布**

$$\boldsymbol{J}_{\mathrm{M}} = \boldsymbol{\nabla}\times\boldsymbol{M} \tag{1.35}$$

**Rmk**　$\rho_{\mathrm{M}} = \boldsymbol{\nabla}\cdot\boldsymbol{J}_{\mathrm{M}} = \boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\times\boldsymbol{M}) \equiv 0.$

**Pf**　$I_{\mathrm{M}} = \int \boldsymbol{J}_{\mathrm{M}}\cdot\mathrm{d}^2\boldsymbol{x} = \oint \underbrace{ni\boldsymbol{S}}_{=n\boldsymbol{m}}\cdot\mathrm{d}\boldsymbol{x} = \oint \boldsymbol{M}\cdot\mathrm{d}\boldsymbol{x} = \int \boldsymbol{\nabla}\times\boldsymbol{M}\cdot\mathrm{d}^2\boldsymbol{x}.$

**Prop. 7　磁化电流的面分布**

$$\boldsymbol{K}_{\mathrm{M}} = \boldsymbol{n}\times(\boldsymbol{M}_2-\boldsymbol{M}_1) = \frac{1}{\mu_0}\boldsymbol{n}\times(\boldsymbol{B}_2-\boldsymbol{B}_1) - \boldsymbol{K} \tag{1.36}$$

**Prop. 8　磁化电流的线分布**

$$I_{\mathrm{M}} = \oint \boldsymbol{M}\cdot\mathrm{d}\boldsymbol{x} \tag{1.37}$$

**Cor. 5 均匀磁化体的总磁矩**

若介质体均匀磁化, 即磁化强度为常矢量 $\boldsymbol{M}$, 其体积为 $V$, 则其总磁矩为

$$\boldsymbol{m}_{\text{total}} = \boldsymbol{M}V \tag{1.38}$$

### 1.2.3 介质中的 Maxwell 方程组

**Def. 7 电位移矢量**

$$\boldsymbol{D} = \varepsilon_0\boldsymbol{E} + \boldsymbol{P} \tag{1.39}$$

**Pf** $\boldsymbol{\nabla}\cdot\boldsymbol{E} = \dfrac{\rho_{\text{total}}}{\varepsilon_0}, \Rightarrow \boldsymbol{\nabla}\cdot\underbrace{\varepsilon_0\boldsymbol{E}+\boldsymbol{P}}_{\overset{\text{def}}{=\!=\!=}} = \rho.$

**Thm. 9 电场关系式**

对于各向同性的均匀线性介质, 假设介质对场的响应是瞬时定域的, 则 $\boldsymbol{P}$ 与 $\boldsymbol{E}$ 有线性关系 $\boldsymbol{P} = \chi_{\text{e}}\varepsilon_0\boldsymbol{E}$, 其中 $\chi_{\text{e}}$ 称为介质的极化率.
设 $\varepsilon = \varepsilon_0(1+\chi_{\text{e}}) = \varepsilon_0\varepsilon_{\text{r}}$, 则 $\boldsymbol{D}$ 与 $\boldsymbol{E}$ 有线性关系

$$\boldsymbol{D} = \varepsilon\boldsymbol{E} \tag{1.40}$$

其中 $\varepsilon$ 称为介电常量, 或电容率; $\varepsilon_{\text{r}}$ 称为相对介电常量, 或相对电容率.

1. 对于非均匀介质, 其介电常量为场点位置的函数

$$\boldsymbol{D}(\boldsymbol{x}) = \varepsilon(\boldsymbol{x})\boldsymbol{E}(\boldsymbol{x}) \tag{1.41}$$

2. 对于各项异性介质, 由介电常量张量 $\varepsilon_{ab}$ 描述介质对外场的响应

$$D_a = \varepsilon_{ab}E^b \tag{1.42}$$

3. 对于非线性介质, $\boldsymbol{D}$ 对 $\boldsymbol{E}$ 非线性依赖

$$D_a = \varepsilon_{ab}E^b + \varepsilon_{abc}E^bE^c + \varepsilon_{abcd}E^bE^cE^d + \cdots \tag{1.43}$$

**Def. 8 磁场强度**

$$\boldsymbol{H} = \frac{1}{\mu_0}\boldsymbol{B} - \boldsymbol{M} \tag{1.44}$$

即 $\boldsymbol{B}=\mu_0\boldsymbol{H}+\mu_0\boldsymbol{M}$. 设 $\boldsymbol{M}=\chi_{\mathrm{m}}\boldsymbol{H}$, $\mu=\mu_0(1+\chi_{\mathrm{m}})$, 则

$$\boldsymbol{B}=\mu\boldsymbol{H} \tag{1.45}$$

**Rmk** 非均匀、各项异性、非线性介质的磁场强度, 可与电位移矢量类似地定义.

**Thm. 10 介质中的 Maxwell 方程组**

$$\begin{cases}\boldsymbol{\nabla}\cdot\boldsymbol{D}=\rho\\ \boldsymbol{\nabla}\times\boldsymbol{E}=-\dfrac{\partial\boldsymbol{B}}{\partial t}\\ \boldsymbol{\nabla}\cdot\boldsymbol{B}=0\\ \boldsymbol{\nabla}\times\boldsymbol{H}=\boldsymbol{J}+\dfrac{\partial\boldsymbol{D}}{\partial t}\end{cases} \tag{1.46}$$

**Pf** $\boldsymbol{\nabla}\cdot\boldsymbol{E}=\dfrac{\rho+\rho_{\mathrm{P}}}{\varepsilon_0}=\dfrac{\rho-\boldsymbol{\nabla}\cdot\boldsymbol{P}}{\varepsilon_0}$, $\Rightarrow\boldsymbol{\nabla}\cdot(\varepsilon_0\boldsymbol{E}+\boldsymbol{P})=\boldsymbol{\nabla}\cdot\boldsymbol{D}=\rho$.

$\boldsymbol{\nabla}\times\boldsymbol{B}=\mu_0(\boldsymbol{J}+\boldsymbol{J}_{\mathrm{P}}+\boldsymbol{J}_{\mathrm{M}})+\mu_0\varepsilon_0\dfrac{\partial\boldsymbol{E}}{\partial t}=\mu_0(\boldsymbol{J}+\boldsymbol{\nabla}\times\boldsymbol{M})+\mu_0\dfrac{\partial}{\partial t}\underbrace{(\varepsilon_0\boldsymbol{E}+\boldsymbol{P})}_{=\boldsymbol{D}}$,

$\Rightarrow\boldsymbol{\nabla}\times\underbrace{(\boldsymbol{B}/\mu_0-\boldsymbol{M})}_{=\boldsymbol{H}}=\boldsymbol{J}+\dfrac{\partial\boldsymbol{D}}{\partial t}$.

**Thm. 11 本构方程**

$$\begin{cases}\boldsymbol{D}=\varepsilon\boldsymbol{E}\\ \boldsymbol{H}=\dfrac{1}{\mu}\boldsymbol{B}\\ \boldsymbol{J}=\sigma\boldsymbol{E}\end{cases} \tag{1.47}$$

## 1.3 电磁场的边值关系

**Lem. 5 积分形式的 Maxwell 方程组**

$$\begin{cases}\displaystyle\oint_{\partial\Omega}\boldsymbol{D}\cdot\mathrm{d}^2\boldsymbol{x}=\int_{\Omega}\rho\,\mathrm{d}^3x\\ \displaystyle\oint_{\partial\Sigma}\boldsymbol{E}\cdot\mathrm{d}\boldsymbol{x}=-\frac{\mathrm{d}}{\mathrm{d}t}\int\boldsymbol{B}_{\Sigma}\cdot\mathrm{d}^2\boldsymbol{x}\\ \displaystyle\oint_{\partial\Omega}\boldsymbol{B}\cdot\mathrm{d}^2\boldsymbol{x}=0\\ \displaystyle\oint_{\partial\Sigma}\boldsymbol{H}\cdot\mathrm{d}\boldsymbol{x}=\int_{\Sigma}\boldsymbol{J}\cdot\mathrm{d}^2\boldsymbol{x}+\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Sigma}\boldsymbol{D}\cdot\mathrm{d}^2\boldsymbol{x}\end{cases} \tag{1.48}$$

**Thm. 12 介质分界面两侧的边值关系**

设 $\boldsymbol{n}$ 为介质 1 指向介质 2 的单位法矢量,

$$\begin{cases} \boldsymbol{n}\cdot(\boldsymbol{D}_2-\boldsymbol{D}_1)=\sigma \\ \boldsymbol{n}\times(\boldsymbol{E}_2-\boldsymbol{E}_1)=0 \\ \boldsymbol{n}\cdot(\boldsymbol{B}_2-\boldsymbol{B}_1)=0 \\ \boldsymbol{n}\times(\boldsymbol{H}_2-\boldsymbol{H}_1)=\boldsymbol{K} \end{cases} \tag{1.49}$$

或可写为

$$\begin{cases} D_{2n}-D_{1n}=\sigma \\ E_{2t}=E_{1t} \\ B_{2n}=B_{1n} \\ H_{2t}-H_{1t}=K \end{cases} \tag{1.50}$$

## 1.4 Maxwell 方程组的对称性 *

### 1.4.1 时空对称性

### 1.4.2 Lorentz 协变性

### 1.4.3 规范对称性

**Def. 9 Lorenz 规范**

$$\nabla\cdot\boldsymbol{A}+\frac{1}{c^2}\frac{\partial\phi}{\partial t}=0 \tag{1.51}$$

**Def. 10 Coulomb 规范**

$$\nabla\cdot\boldsymbol{A}=0 \tag{1.52}$$

### 1.4.4 分立对称性

## 1.5 Lorentz 力

**Def. 11 Lorentz 力密度**

$$f=\rho\boldsymbol{E}+\boldsymbol{J}\times\boldsymbol{B} \tag{1.53}$$

**Def. 12 Lorentz 力**

带电荷量 $q$, 以速度 $\boldsymbol{v}$ 运动的带电粒子在电磁场中所受力为

$$\boldsymbol{F} = q\boldsymbol{E} + q\boldsymbol{v} \times \boldsymbol{B} \tag{1.54}$$

# 2 电磁场的守恒律

## 2.1 电磁场能量

### 2.1.1 能量守恒定律

**Law. 1 开放体系的能量守恒定律**

$$\oint_{\partial\Omega} \boldsymbol{S}\cdot \mathrm{d}^2\boldsymbol{x} + \frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega} w\,\mathrm{d}^3x + \int_{\Omega} \boldsymbol{f}\cdot\boldsymbol{v}\,\mathrm{d}^3x = 0 \tag{2.1}$$

**Thm. 1 能量守恒律的微分形式**

$$\boldsymbol{\nabla}\cdot\boldsymbol{S} + \frac{\partial w}{\partial t} + \boldsymbol{f}\cdot\boldsymbol{v} = 0 \tag{2.2}$$

**Cor. 1 孤立系统或全空间的能量守恒律**

对于封闭体系 $\Omega$ 或全空间, 边界面上无能量流动, 即

$$\oint_{\partial\Omega/\infty} \boldsymbol{S}\cdot\mathrm{d}^2\boldsymbol{x} = 0 \tag{2.3}$$

电磁场能量与带电体的机械能可相互转化

$$\int_{\Omega/\infty} \boldsymbol{f}\cdot\boldsymbol{v}\,\mathrm{d}^3x = -\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega/\infty} w\,\mathrm{d}^3x \tag{2.4}$$

### 2.1.2 能量密度与能流密度

**Prop. 1 能流密度与能量密度的表达式**

对于一般介质, 能流密度 (Poynting 矢量) 与能量密度的时间变化率为

$$\boldsymbol{S} = \boldsymbol{E}\times\boldsymbol{H} \tag{2.5}$$

$$\frac{\partial w}{\partial t} = \boldsymbol{E}\cdot\frac{\partial \boldsymbol{D}}{\partial t} + \boldsymbol{H}\cdot\frac{\partial \boldsymbol{B}}{\partial t} \tag{2.6}$$

对于各向同性的非色散线性介质, 能量密度为

$$w = \frac{1}{2}(\boldsymbol{E}\cdot\boldsymbol{D} + \boldsymbol{H}\cdot\boldsymbol{B}) \tag{2.7}$$

另对于各向同性的均匀介质

$$\boldsymbol{S} = \frac{1}{\mu}\boldsymbol{E}\times\boldsymbol{B} \qquad w = \frac{1}{2}\left(\varepsilon E^2 + \frac{1}{\mu}B^2\right) \tag{2.8}$$

**Pf** $\displaystyle\int \boldsymbol{f}\cdot\boldsymbol{v}\,\mathrm{d}^3x=\int \boldsymbol{J}\cdot\boldsymbol{E}\,\mathrm{d}^3x,\ \boldsymbol{J}\cdot\boldsymbol{E}=\left(\nabla\times\boldsymbol{H}+\frac{\partial\boldsymbol{D}}{\partial t}\right)\cdot\boldsymbol{E}=\boldsymbol{E}\cdot(\nabla\times\boldsymbol{H})-\boldsymbol{E}\cdot\frac{\partial\boldsymbol{D}}{\partial t}.$

$\displaystyle\boldsymbol{E}\cdot(\nabla\times\boldsymbol{H})=\boldsymbol{H}\cdot(\nabla\times\boldsymbol{E})-\nabla\cdot(\boldsymbol{E}\times\boldsymbol{H})=-\boldsymbol{H}\cdot\frac{\partial\boldsymbol{B}}{\partial t}-\nabla\cdot(\boldsymbol{E}\times\boldsymbol{H}).$

若介质线性非色散, 则有 $\displaystyle\boldsymbol{E}\cdot\frac{\partial\boldsymbol{D}}{\partial t}=\frac{\partial}{\partial t}\frac{\boldsymbol{E}\cdot\boldsymbol{D}}{2},\ \boldsymbol{H}\cdot\frac{\partial\boldsymbol{B}}{\partial t}=\frac{\partial}{\partial t}\frac{\boldsymbol{H}\cdot\boldsymbol{B}}{2}.$

$$\Rightarrow \boldsymbol{f}\cdot\boldsymbol{v}=-\nabla\cdot\underbrace{(\boldsymbol{E}\times\boldsymbol{H})}_{\overset{\text{def}}{=\!=}\boldsymbol{S}}-\frac{\partial}{\partial t}\underbrace{\frac{\boldsymbol{E}\cdot\boldsymbol{D}+\boldsymbol{H}\cdot\boldsymbol{B}}{2}}_{\overset{\text{def}}{=\!=}w}$$

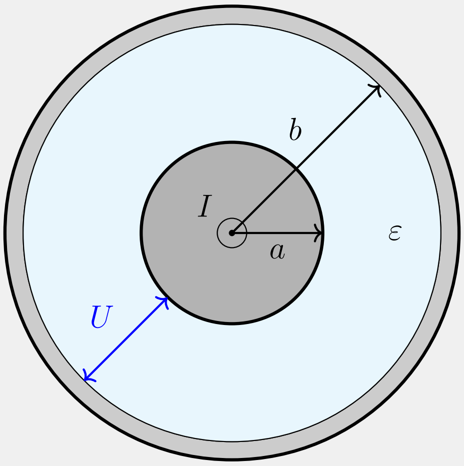
### 2.1.3 电磁能量的传输

**Eg. 1 电容器的能量流动**

**Eg. 2 同轴电缆的能量流动**

$$\boldsymbol{S}=\frac{UI}{2\pi r^2\ln(b/a)}\boldsymbol{e}_z \tag{2.9}$$

$$S_r=-\frac{I^2}{2\pi^2a^3\sigma} \tag{2.10}$$




## 2.2 电磁场动量

### 2.2.1 动量守恒定律

**Def. 1 Minkowski 力密度**

$$\boldsymbol{f}_{\mathrm{M}}=\rho\boldsymbol{E}+\boldsymbol{J}\times\boldsymbol{B}-\frac{1}{2}E^2\nabla\varepsilon-\frac{1}{2}B^2\nabla\mu \tag{2.11}$$

**Thm. 2 微分形式的动量定理**

$$\boldsymbol{f}_{\mathrm{M}}+\frac{\partial \boldsymbol{g}_{\mathrm{M}}}{\partial t}=-\boldsymbol{\nabla}\cdot\mathcal{T} \tag{2.12}$$

**Pf** $\boldsymbol{f}=\rho\boldsymbol{E}+\boldsymbol{J}\times\boldsymbol{B}=(\boldsymbol{\nabla}\cdot\boldsymbol{D})\boldsymbol{E}+(\boldsymbol{\nabla}\times\boldsymbol{H})\times\boldsymbol{B}-\dfrac{\partial\boldsymbol{D}}{\partial t}\times\boldsymbol{B}$. 由 $\boldsymbol{\nabla}\cdot\boldsymbol{B}=0$, $\boldsymbol{\nabla}\times\boldsymbol{E}=-\dfrac{\partial\boldsymbol{B}}{\partial t}$, 有 $\boldsymbol{f}=(\boldsymbol{\nabla}\cdot\boldsymbol{D})\boldsymbol{E}+(\boldsymbol{\nabla}\cdot\boldsymbol{B})\boldsymbol{H}-\dfrac{\partial}{\partial t}(\boldsymbol{D}\times\boldsymbol{B})-\boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{H})-\boldsymbol{D}\times(\boldsymbol{\nabla}\times\boldsymbol{E})$.

由 $\begin{cases}\boldsymbol{B}\times(\boldsymbol{\nabla}\times\boldsymbol{H})=(\boldsymbol{\nabla}\boldsymbol{H})\cdot\boldsymbol{B}-(\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{H}\\ \boldsymbol{D}\times(\boldsymbol{\nabla}\times\boldsymbol{E})=(\boldsymbol{\nabla}\boldsymbol{E})\cdot\boldsymbol{D}-(\boldsymbol{D}\cdot\boldsymbol{\nabla})\boldsymbol{E}\end{cases}$ 与 $\begin{cases}(\boldsymbol{\nabla}\cdot\boldsymbol{D})\boldsymbol{E}+(\boldsymbol{D}\cdot\boldsymbol{\nabla})\boldsymbol{E}=\boldsymbol{\nabla}\cdot(\boldsymbol{D}\boldsymbol{E})\\ (\boldsymbol{\nabla}\cdot\boldsymbol{B})\boldsymbol{H}+(\boldsymbol{B}\cdot\boldsymbol{\nabla})\boldsymbol{H}=\boldsymbol{\nabla}\cdot(\boldsymbol{B}\boldsymbol{H})\end{cases}$

$$\boldsymbol{f}=\boldsymbol{\nabla}\cdot(\boldsymbol{D}\boldsymbol{E}+\boldsymbol{B}\boldsymbol{H})-\frac{\partial}{\partial t}(\boldsymbol{D}\times\boldsymbol{B})-(\boldsymbol{\nabla}\boldsymbol{H})\cdot\boldsymbol{B}-(\boldsymbol{\nabla}\boldsymbol{E})\cdot\boldsymbol{D}$$

由 $\boldsymbol{\nabla}(\boldsymbol{D}\cdot\boldsymbol{E})=(\boldsymbol{\nabla}\boldsymbol{D})\cdot\boldsymbol{E}+(\boldsymbol{\nabla}\boldsymbol{E})\cdot\boldsymbol{D}$, $\boldsymbol{\nabla}\boldsymbol{D}\cdot\boldsymbol{E}=\boldsymbol{\nabla}(\varepsilon\boldsymbol{E})\cdot\boldsymbol{E}=E^2\boldsymbol{\nabla}\varepsilon+(\boldsymbol{\nabla}\boldsymbol{E})\cdot\boldsymbol{D}$, 有 $(\boldsymbol{\nabla}\boldsymbol{E})\cdot\boldsymbol{D}=\dfrac{1}{2}\left[\boldsymbol{\nabla}(\boldsymbol{D}\cdot\boldsymbol{E})-E^2\boldsymbol{\nabla}\varepsilon\right]$, 同理 $(\boldsymbol{\nabla}\boldsymbol{H})\cdot\boldsymbol{B}=\dfrac{1}{2}\left[\boldsymbol{\nabla}(\boldsymbol{B}\cdot\boldsymbol{H})-H^2\boldsymbol{\nabla}\mu\right]$.

又由 $\boldsymbol{\nabla}\phi=\boldsymbol{\nabla}\cdot(\phi\mathcal{I})$, 可得

$$\underbrace{\boldsymbol{f}-\frac{1}{2}E^2\boldsymbol{\nabla}\varepsilon-\frac{1}{2}H^2\boldsymbol{\nabla}\mu}_{\overset{\text{def}}{=\!=}\boldsymbol{f}_{\mathrm{M}}}=-\boldsymbol{\nabla}\cdot\underbrace{\left[\frac{1}{2}(\boldsymbol{D}\cdot\boldsymbol{E}+\boldsymbol{B}\cdot\boldsymbol{H})\mathcal{I}-\boldsymbol{D}\boldsymbol{E}-\boldsymbol{B}\boldsymbol{H}\right]}_{\overset{\text{def}}{=\!=}\mathcal{T}}-\frac{\partial}{\partial t}\underbrace{(\boldsymbol{D}\times\boldsymbol{B})}_{\overset{\text{def}}{=\!=}\boldsymbol{g}_{\mathrm{M}}}$$

**Law. 2 开放体系的动量守恒定律**

对于开放体系 $\Omega$, 带电体系与电磁场的总动量变化率等于从 $\Omega$ 外通过边界 $\partial\Omega$ 流入的动量, 即

$$\frac{\mathrm{d}\boldsymbol{G}_{\mathrm{mech}}}{\mathrm{d}t}+\frac{\mathrm{d}\boldsymbol{G}_{\mathrm{em}}}{\mathrm{d}t}=\int_{\Omega}\boldsymbol{f}_{\mathrm{M}}\,\mathrm{d}^3x+\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega}\boldsymbol{g}_{\mathrm{M}}\,\mathrm{d}^3x=-\oint_{\partial\Omega}\boldsymbol{n}\cdot\mathcal{T}\,\mathrm{d}^2x \tag{2.13}$$

**Cor. 2 封闭体系的动量守恒定律**

对于封闭体系 $\bar{\Omega}$ 或全空间 $\infty$, 带电体系的机械动量与电磁场动量守恒, 即

$$\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{G}_{\mathrm{mech}}+\boldsymbol{G}_{\mathrm{em}})=\int_{\bar{\Omega}/\infty}\boldsymbol{f}_{\mathrm{M}}\,\mathrm{d}^3x+\frac{\mathrm{d}}{\mathrm{d}t}\int_{\bar{\Omega}/\infty}\boldsymbol{g}_{\mathrm{M}}\,\mathrm{d}^3x=0 \tag{2.14}$$

$$\Longrightarrow \boldsymbol{G}_{\mathrm{mech}}+\boldsymbol{G}_{\mathrm{em}}=\mathrm{Const} \tag{2.15}$$

### 2.2.2 动量密度

**Def. 2 动量密度**

电磁场的动量密度矢量为

$$\boldsymbol{g}_{\mathrm{M}} = \boldsymbol{D} \times \boldsymbol{B} \tag{2.16}$$

其体积分即为区域 $\Omega$ 内的电磁场动量

$$\boldsymbol{G}_{\mathrm{em}} = \int \boldsymbol{g}\, \mathrm{d}^3 x \tag{2.17}$$

**Prop. 2**

动量密度与能流密度方向相同.

$$\boldsymbol{g}_{\mathrm{M}} = \mu\varepsilon \boldsymbol{E} \times \boldsymbol{H} = \frac{\mu_{\mathrm{r}}\varepsilon_{\mathrm{r}}}{c^2}\boldsymbol{S} \tag{2.18}$$

### 2.2.3 动量流密度 (Maxwell 应力张量)

**Def. 3 动量流密度**

$$\mathcal{T} = -\boldsymbol{DE} - \boldsymbol{BH} + \frac{1}{2}(\boldsymbol{D}\cdot\boldsymbol{E} + \boldsymbol{B}\cdot\boldsymbol{H})\mathcal{I} \tag{2.19}$$

$$T_{ij} = -\varepsilon E_i E_j - \frac{1}{\mu}B_i B_j + \frac{1}{2}\left(\varepsilon\delta_{ij}E^2 + \frac{1}{\mu}\delta_{ij}B^2\right) \tag{2.20}$$

**Prop. 3 光压**

## 2.3 电磁场角动量

### 2.3.1 角动量守恒定律

**Lem. 1**

$$\boldsymbol{x} \times (\boldsymbol{\nabla}\cdot\mathcal{T}) = -\boldsymbol{\nabla}\cdot(\mathcal{T} \times \boldsymbol{x}) \tag{2.21}$$

**Pf** $\boldsymbol{x} \times (\boldsymbol{\nabla}\cdot\mathcal{T}) = \varepsilon_{ijk}x_i(\boldsymbol{\nabla}\cdot\mathcal{T})_j\boldsymbol{e}_k = \varepsilon_{ijk}x_i\partial_l T_{lj}\boldsymbol{e}_k = \varepsilon_{ijk}\partial_l(x_i T_{lj})\boldsymbol{e}_k - \underbrace{\varepsilon_{ijk}T_{lj}\delta_{il}\boldsymbol{e}_k}_{=0}.$

$-\boldsymbol{\nabla}\cdot(\mathcal{T} \times \boldsymbol{x}) = -\boldsymbol{\nabla}\cdot(T_{lj}\boldsymbol{e}_l\boldsymbol{e}_j \times x_i\boldsymbol{e}_i) = -\boldsymbol{\nabla}\cdot(\boldsymbol{e}_l x_i T_{lj}\varepsilon_{jik}\boldsymbol{e}_k) = \varepsilon_{ijk}\partial_l(x_i T_{lj})\boldsymbol{e}_k.$

**Thm. 3 微分形式的角动量定理**

$$\boldsymbol{x} \times \boldsymbol{f}_{\mathrm{M}} + \frac{\partial \boldsymbol{l}_{\mathrm{em}}}{\partial t} = -\boldsymbol{\nabla}\cdot\mathcal{M}_{\mathrm{em}} \tag{2.22}$$

**Pf** 由 $\boldsymbol{f}_{\mathrm{M}}+\dfrac{\partial \boldsymbol{g}_{\mathrm{M}}}{\partial t}=-\boldsymbol{\nabla}\cdot\mathcal{T}$ 可得 $\boldsymbol{x}\times\boldsymbol{f}_{\mathrm{M}}+\boldsymbol{x}\times\dfrac{\partial \boldsymbol{g}_{\mathrm{M}}}{\partial t}=-\boldsymbol{x}\times(\boldsymbol{\nabla}\cdot\mathcal{T})$.

$$\boldsymbol{x}\times\frac{\partial \boldsymbol{g}_{\mathrm{M}}}{\partial t}=\frac{\partial}{\partial t}(\boldsymbol{x}\times\boldsymbol{g}_{\mathrm{M}})-\underbrace{\frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t}\times\boldsymbol{g}_{\mathrm{M}}}_{=0}\overset{\text{def}}{=\!=}\frac{\partial \boldsymbol{l}_{\mathrm{em}}}{\partial t}$$

由引理, $-\boldsymbol{x}\times(\boldsymbol{\nabla}\cdot\mathcal{T})=-\boldsymbol{\nabla}\cdot(-\mathcal{T}\times\boldsymbol{x})\overset{\text{def}}{=\!=}-\boldsymbol{\nabla}\cdot\mathcal{M}_{\mathrm{em}}$.

**Law. 3 开放体系的角动量守恒定律**

$$\frac{\mathrm{d}\boldsymbol{L}_{\mathrm{mech}}}{\mathrm{d}t}+\frac{\mathrm{d}\boldsymbol{L}_{\mathrm{em}}}{\mathrm{d}t}=\int_{\Omega}\boldsymbol{x}\times\boldsymbol{f}_{\mathrm{M}}\,\mathrm{d}^3x+\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega}\boldsymbol{l}_{\mathrm{em}}\,\mathrm{d}^3x=-\oint_{\partial\Omega}\boldsymbol{n}\cdot\mathcal{M}_{\mathrm{em}}\,\mathrm{d}^2x \tag{2.23}$$

**Cor. 3 封闭体系的角动量守恒定律**

对于封闭体系 $\bar{\Omega}$ 或全空间 $\infty$, 带电体系的机械角动量与电磁场角动量守恒, 即

$$\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{L}_{\mathrm{mech}}+\boldsymbol{L}_{\mathrm{em}})=\int_{\bar{\Omega}/\infty}\boldsymbol{x}\times\boldsymbol{f}_{\mathrm{M}}\,\mathrm{d}^3x+\frac{\mathrm{d}}{\mathrm{d}t}\int_{\bar{\Omega}/\infty}\boldsymbol{l}_{\mathrm{em}}\,\mathrm{d}^3x=0 \tag{2.24}$$

$$\Longrightarrow \boldsymbol{L}_{\mathrm{mech}}+\boldsymbol{L}_{\mathrm{em}}=\mathrm{Const} \tag{2.25}$$

### 2.3.2 角动量密度

**Def. 4 角动量密度**

$$\boldsymbol{l}_{\mathrm{em}}=\boldsymbol{x}\times\boldsymbol{g}_{\mathrm{M}} \tag{2.26}$$

### 2.3.3 角动量流密度

**Def. 5 角动量流密度**

$$\mathcal{M}_{\mathrm{em}}=-\mathcal{T}\times\boldsymbol{x} \tag{2.27}$$

# 3 静电学

**Lem. 1 静电场基本方程**

$$\begin{cases} \boldsymbol{\nabla} \cdot \boldsymbol{D} = \rho \\ \boldsymbol{\nabla} \times \boldsymbol{E} = 0 \\ \boldsymbol{D} = \varepsilon \boldsymbol{E} \\ \begin{cases} \boldsymbol{n} \cdot (\boldsymbol{D}_2 - \boldsymbol{D}_1) = \sigma \\ \boldsymbol{n} \times (\boldsymbol{E}_2 - \boldsymbol{E}_1) = 0 \end{cases} \end{cases} \tag{3.1}$$

## 3.1 静电势

### 3.1.1 静电势的微分方程

**Prop. 1 电场强度的静电势表示**

$$\boldsymbol{E} = -\boldsymbol{\nabla}\phi \tag{3.2}$$

$$E_n = -\frac{\partial \phi}{\partial n} \tag{3.3}$$

$$\mathrm{d}\phi = -\boldsymbol{E} \cdot \mathrm{d}\boldsymbol{x} \tag{3.4}$$

**Cor. 1 静电势的求解**

设 $\phi(p_0) = 0$, 即取 $p_0$ 为参考点, 则

$$\phi(p) = -\int_{p_0}^{p} \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{x} \tag{3.5}$$

**Eg. 1 点电荷的静电势**

$$\phi(r) = \frac{q}{4\pi\varepsilon_0 r} \tag{3.6}$$

**Eg. 2 均匀电场的静电势**

设均匀电场 $\boldsymbol{E}_0$ 的原点为参考点, 则

$$\phi(\boldsymbol{x}) = -\boldsymbol{E}_0 \cdot \boldsymbol{x} = -E_0 x \cos\theta \tag{3.7}$$

**Lem. 2**

$$\int \frac{\mathrm{d}z}{\sqrt{z^2+R^2}} = \ln\left(z+\sqrt{z^2+R^2}\right)+C \tag{3.8}$$

**Eg. 3 无限长带电直线的静电势**

设均匀带电无限长直线的电荷线密度为 $\lambda$, 则距离 $R$ 远处的电势为

$$\phi(R) = -\frac{\lambda}{2\pi\varepsilon_0}\ln\frac{R}{R_0} \tag{3.9}$$

**Prop. 2 Poisson 方程**

$$\nabla^2\phi(\boldsymbol{x}) = -\frac{\rho(\boldsymbol{x})}{\varepsilon} \tag{3.10}$$

对于存在自由电荷体分布的区域成立.

**Prop. 3 Laplace 方程**

$$\nabla^2\phi(\boldsymbol{x}) = 0 \tag{3.11}$$

对于无自由电荷体分布的区域成立.

### 3.1.2 边值关系

**Thm. 1 介质分界面的边值关系**

$$\begin{cases} \phi_2 = \phi_1 \\ \varepsilon_2\dfrac{\partial\phi_2}{\partial n} - \varepsilon_1\dfrac{\partial\phi_1}{\partial n} = -\sigma \end{cases} \tag{3.12}$$

**Pf** $\phi_2-\phi_1 = -\int_{p_1}^{p_2}\boldsymbol{E}\cdot\mathrm{d}\boldsymbol{x}\to 0$, 因此介质分界面处电势连续, 有 $\phi_2=\phi_1$. 设介质分界面存在自由电荷分布 $\sigma$, 则 $D_{2n}-D_{1n}=\varepsilon_2E_{2n}-\varepsilon_1E_{1n} = -\left(\varepsilon_2\dfrac{\partial\phi_2}{\partial n}-\varepsilon_1\dfrac{\partial\phi_1}{\partial n}\right)=\sigma$.

**Thm. 2 绝缘介质分界面的边值关系**

$$\begin{cases} \phi_2 = \phi_1 \\ \varepsilon_2\dfrac{\partial\phi_2}{\partial n} = \varepsilon_1\dfrac{\partial\phi_1}{\partial n} \end{cases} \tag{3.13}$$

**Pf** 绝缘介质分界面一般无自由电荷分布, 即 $\varepsilon_2E_{2n}=\varepsilon_1E_{1n}$, 因此 $\varepsilon_2\dfrac{\partial\phi_2}{\partial n}=\varepsilon_1\dfrac{\partial\phi_1}{\partial n}$.

**Thm. 3 导体表面边值关系**

设导体表面自由电荷分布为 $\sigma$, 外侧介质的介电常量为 $\varepsilon$, 则边值关系为

$$\begin{cases} \phi|_{\text{surf}} = \text{Const} \\ \dfrac{\partial \phi}{\partial n} = -\dfrac{\sigma}{\varepsilon} \end{cases} \tag{3.14}$$

**Pf** 由 $\boldsymbol{n} \times \boldsymbol{E} = 0$ 与 $\boldsymbol{n} \cdot \boldsymbol{D} = \sigma$ 易得.

**Prop. 4 导体表面自由电荷量**

若导体表面周围介质均匀, 整个导体表面总自由电荷量为

$$Q_{\text{f}} = \oint_{\text{surf}} \sigma \, \mathrm{d}^2 x = -\varepsilon \oint_{\text{surf}} \frac{\partial \phi}{\partial n} \, \mathrm{d}^2 x \tag{3.15}$$

**Eg. 4 带电导体球壳的边界条件**

设导体球壳带电 $Q$, 内外半径分别为 $R_1, R_2$, 壳内外电势分别为 $\phi_1, \phi_2$, 则

$$-\oint_{r=R_2} \frac{\partial \phi_2}{\partial r} \, \mathrm{d}^2 x + \oint_{r=R_1} \frac{\partial \phi_1}{\partial r} \, \mathrm{d}^2 x = \frac{Q}{\varepsilon_0} \tag{3.16}$$

**Cor. 2 导体的静电条件**

1. 内部无自由电荷, 电荷只分布于表面;
2. 内部电场为零;
3. 整个导体表面为等势面, 整个导体为等势体.

### 3.1.3 静电场的能量

**Prop. 5 静电体系总能量**

$$W_\infty = \frac{1}{2} \int_\infty \rho(\boldsymbol{x}) \phi(\boldsymbol{x}) \, \mathrm{d}^3 x \tag{3.17}$$

**Rmk** 局部电场能量为 $W_\Omega = \dfrac{1}{2} \displaystyle\int_\Omega \boldsymbol{E} \cdot \boldsymbol{D} \, \mathrm{d}^3 x$, 能量密度 $w_{\text{e}} = \dfrac{\boldsymbol{E} \cdot \boldsymbol{D}}{2} \neq \dfrac{\rho \phi}{2}$.

**Pf** $W_\Omega = \dfrac{1}{2} \displaystyle\int_\Omega \boldsymbol{E} \cdot \boldsymbol{D} \, \mathrm{d}^3 x = -\dfrac{1}{2} \displaystyle\int_\Omega \boldsymbol{\nabla} \phi \cdot \boldsymbol{D} \, \mathrm{d}^3 x = \dfrac{1}{2} \displaystyle\int_\Omega \rho \phi \, \mathrm{d}^3 x - \dfrac{1}{2} \displaystyle\int_{\partial \Omega} \phi \boldsymbol{D} \cdot \mathrm{d}^2 \boldsymbol{x}$, 当 $\Omega \to \infty$, $\phi D \sim r^{-3}$, $\partial \Omega \sim r^2$, $\Rightarrow W_\infty = \dfrac{1}{2} \displaystyle\int_\infty \rho \phi \, \mathrm{d}^3 x$.

**Cor. 3 导体总能量**

由于导体表面等势, 且电荷只分布于表面, 有

$$W_{\text{cond}} = \frac{1}{2} Q\phi \tag{3.18}$$

**Lem. 3 Green 互易定理**

$$\int \rho_1 \phi_2 \, \mathrm{d}^3 x = \int \rho_2 \phi_1 \, \mathrm{d}^3 x \tag{3.19}$$

**Prop. 6 电荷体系间的相互作用能**

$$W_{\text{int}} = \int_{\Omega'} \rho \phi_{\text{ext}} \, \mathrm{d}^3 x' \tag{3.20}$$

**Pf** 设 $\rho, \rho_{\text{ext}}$ 产生的静电势分别为 $\phi, \phi_{\text{ext}}$. $W_{\text{total}} = \dfrac{1}{2} \displaystyle\int_{\Omega' + \Omega'_{\text{ext}}} (\rho + \rho_{\text{ext}})(\phi + \phi_{\text{ext}}) \, \mathrm{d}^3 x'$, 展开得 $W_{\text{total}} = \underbrace{\dfrac{1}{2} \displaystyle\int_{\Omega'} \rho\phi \, \mathrm{d}^3 x' + \dfrac{1}{2} \int_{\Omega'_{\text{ext}}} \rho_{\text{ext}} \phi_{\text{ext}} \, \mathrm{d}^3 x'}_{= W_{\text{self}}} + \underbrace{\dfrac{1}{2} \displaystyle\int_{\Omega'} \rho \phi_{\text{ext}} \, \mathrm{d}^3 x' + \dfrac{1}{2} \int_{\Omega'_{\text{ext}}} \rho_{\text{ext}} \phi \, \mathrm{d}^3 x'}_{= W_{\text{int}}}$, 结合 Green 互易定理得证.

### 3.1.4 Earnshaw 定理

**Thm. 4 Earnshaw 定理**

仅靠静电力, 无法将粒子维持在稳定的力学平衡状态. 等价地, 静电势在有限无自由电荷分布的区域内, 不能达到极值.

## 3.2 静电场唯一性定理 *

**Thm. 5 静电场唯一性定理**

已知区域 $\Omega$ 内的介质介电常量分布 $\varepsilon(\boldsymbol{x})$, 若给定

- $\Omega$ 内的自由电荷分布 $\rho(\boldsymbol{x})$;
- 边界 $\partial\Omega$ 上的
  - 电势 $\phi|_{\partial\Omega}$,
  - 电势的法向导数 $\left.\dfrac{\partial \phi}{\partial n}\right|_{\partial\Omega}$,

则 $\Omega$ 内的静电场唯一确定, 或 $\Omega$ 内 Poisson 方程的解在相差一个常量的意义下唯一.

## 3.3 镜像法

**Meth. 1 镜像法**

1. 使用条件
   (a) 求解区域的边界具有一定的对称性.
   (b) 求解区域内的自由电荷少量, 其电势 $\phi$ 可直接给出解析解.
   (c) 边界上感应面电荷或极化面电荷, 对求解区域的电势产生的效果, 可用镜像电荷代替.
2. 在求解区域外引入镜像电荷, 则求解区域的总电势为 $\phi_{\text{total}} = \phi + \phi'$.
3. 调整镜像电荷的位置与电荷量, 使总电势满足边界条件, 则根据静电场唯一性定理, $\phi_{\text{total}}$ 即所求 Poisson 方程的解;
4. 可进一步利用镜像电荷分析求解区域的 $\boldsymbol{E}$ 与 $\boldsymbol{F}$, 边界上的 $\sigma$.

**Eg. 5 无限大接地导体平面**

点电荷 $q$ 距离无限大接地导体平面为 $d$, 设其位置为 $(0,0,d)$, 则镜像电荷 $q' = -q$, 位置为 $(0,0,-d)$. 求解域的电势为

$$\phi(x,y,z) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{\sqrt{x^2+y^2+(z-d)^2}} - \frac{q}{\sqrt{x^2+y^2+(z+d)^2}}\right] \tag{3.21}$$

感应电荷面密度为

$$\sigma(x,y) = -\varepsilon_0 \left.\frac{\partial\phi}{\partial z}\right|_{z=0} = -\frac{qd}{2\pi(x^2+y^2+d^2)^{3/2}} \tag{3.22}$$

总感应电荷

$$Q = \int_{-\infty}^{\infty} \sigma(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_{r=0}^{\infty}\int_{\theta=0}^{2\pi} \sigma(r,\theta)\,\mathrm{d}r\,\mathrm{d}\theta = -q = q' \tag{3.23}$$

点电荷受力为

$$\boldsymbol{F}(z) = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{(2z)^2}\boldsymbol{e}_z \tag{3.24}$$

**Eg. 6 接地导体球**

点电荷 $q$ 距离半径为 $R$ 的接地导体球球心为 $d(>R)$, 则镜像电荷 $q' = -\dfrac{R}{d}q$, 距离球心为

$d' = \dfrac{R^2}{d}$. 求解域电势为

$$\phi(r,\theta) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{\sqrt{r^2+d^2-2rd\cos\theta}} - \frac{\frac{R}{d}q}{\sqrt{r^2+\left(\frac{R^2}{d}\right)^2-2r\frac{R^2}{d}\cos\theta}}\right] \tag{3.25}$$

若导体球不接地, 而带电荷 $Q_0$, 则感应电荷可等效为两个镜像电荷的叠加:

1. $q' = -\dfrac{R}{d}q$, 距离球心为 $d' = \dfrac{R^2}{d}$;
2. $q'' = Q_0 - q'$, 位于球心.

求解域电势为

$$\phi(r,\theta) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{\sqrt{r^2+d^2-2rd\cos\theta}} - \frac{\frac{R}{d}q}{\sqrt{r^2+\left(\frac{R^2}{d}\right)^2-2r\frac{R^2}{d}\cos\theta}} + \frac{Q_0+\frac{R}{d}Q}{r}\right] \tag{3.26}$$

**Eg. 7 组合接地边界**

组合边界通常由平面, 球面, 柱面 (可视为二维球面) 的部分组合而成, 需进行多次镜像.

1. 球面或柱面 + 球面或柱面: 通常不用镜像法求解.
2. 两相交平面: 夹角为 $\pi/n$ 且电荷位于角平分面上时可解.
3. 平面 + 球面或柱面: 球心或圆柱轴线位于平面上时可解. 先对球面或柱面做镜像 $q'$, 再将 $q$ 与第一次的镜像电荷 $q'$ 对平面做镜像, 得到 $-q$ 和 $-q'$.

## 3.4 Green 函数法

### 3.4.1 解的 Green 函数表示

**Def. 1 Green 函数**

Green 函数为单位点源影响函数

$$\nabla^2 G(\boldsymbol{x},\boldsymbol{x}') = -\frac{\delta(\boldsymbol{x}-\boldsymbol{x}')}{\varepsilon} \tag{3.27}$$

满足第一类或第二类边界条件

$$G_{\mathrm{D}}(\boldsymbol{x},\boldsymbol{x}')|_S = 0 \tag{3.28}$$

$$\left.\frac{\partial G_{\mathrm{N}}(\boldsymbol{x},\boldsymbol{x}')}{\partial n}\right|_S = -\frac{1}{\varepsilon S} \tag{3.29}$$

**Lem. 4 第一 Green 公式**

$$\int_\Omega \phi\nabla^2\psi + \boldsymbol{\nabla}\phi\cdot\boldsymbol{\nabla}\psi\,\mathrm{d}^3x = \oint_{\partial\Omega}\phi\boldsymbol{\nabla}\psi\cdot\mathrm{d}^2\boldsymbol{x} \tag{3.30}$$

**Pf** $\boldsymbol{\nabla}\cdot(\phi\boldsymbol{\nabla}\psi) = \boldsymbol{\nabla}\phi\cdot\boldsymbol{\nabla}\psi + \phi\nabla^2\psi$.

**Lem. 5 第二 Green 公式**

$$\int_\Omega \phi\nabla^2\psi - \psi\nabla^2\phi\,\mathrm{d}^3x = \oint_{\partial\Omega}(\phi\boldsymbol{\nabla}\psi - \psi\boldsymbol{\nabla}\phi)\cdot\mathrm{d}^2\boldsymbol{x} \tag{3.31}$$

**Pf** $\boldsymbol{\nabla}\cdot(\phi\boldsymbol{\nabla}\psi - \psi\boldsymbol{\nabla}\phi) = (\boldsymbol{\nabla}\phi\cdot\boldsymbol{\nabla}\psi + \phi\nabla^2\psi) - (\boldsymbol{\nabla}\psi\cdot\boldsymbol{\nabla}\phi + \psi\nabla^2\phi) = \phi\nabla^2\psi - \psi\nabla^2\phi$.

**Prop. 7 解的表示**

对于各向同性的线性均匀介质, Poisson 方程的解为

$$\phi(\boldsymbol{x}) = \int_{\Omega'}\rho(\boldsymbol{x}')G(\boldsymbol{x}',\boldsymbol{x})\,\mathrm{d}^3x' + \varepsilon\oint_{\partial\Omega'} G\frac{\partial\phi}{\partial n'} - \phi\frac{\partial G}{\partial n'}\,\mathrm{d}^2x' \tag{3.32}$$

**Rmk** 选取 $G(\boldsymbol{x}',\boldsymbol{x})$ 的边界条件, 可使得 $\phi(\boldsymbol{x})$ 只与 $\phi(\boldsymbol{x}')$, $\dfrac{\partial\phi(\boldsymbol{x}')}{\partial n'}$ 其一相关.

**Pf** 第二 Green 公式中令 $\psi = G$.

$$\Rightarrow \int_\Omega \phi\nabla^2 G - G\nabla^2\phi\,\mathrm{d}^3x = \int_\Omega \frac{G(\boldsymbol{x},\boldsymbol{x}')\rho(\boldsymbol{x})}{\varepsilon} - \frac{\phi(\boldsymbol{x})\delta(\boldsymbol{r})}{\varepsilon}\,\mathrm{d}^3x = \frac{1}{\varepsilon}\int_\Omega \rho(\boldsymbol{x})G(\boldsymbol{x},\boldsymbol{x}')\,\mathrm{d}^3x - \frac{\phi(\boldsymbol{x}')}{\varepsilon}$$

$$\Rightarrow \phi(\boldsymbol{x}') = \int_\Omega \rho(\boldsymbol{x})G(\boldsymbol{x},\boldsymbol{x}')\,\mathrm{d}^3x + \varepsilon\oint_{\partial\Omega} G\frac{\partial\phi}{\partial n} - \phi\frac{\partial G}{\partial n}\,\mathrm{d}^2x$$

做场源互换 $\boldsymbol{x}\leftrightarrow\boldsymbol{x}'$, 由 Green 函数对场点与源点的对称性, 得证.

### 3.4.2 Dirichlet 边值问题

### 3.4.3 Neumann 边界问题

### 3.4.4 几类特殊边值问题的 Green 函数

## 3.5 Laplace 方程的分离变量法

**Meth. 2 Poisson 方程问题的简化**

若已知求解域内的电荷产生的电势为 $\phi_{\mathrm{f}}$, 则可令 Poisson 方程的解为

$$\phi = \phi_{\mathrm{f}} + \phi' \tag{3.33}$$

其中 $\phi'$ 满足 Laplace 方程. 可用分离变量法求解 $\phi'$, 再利用解的叠加原理得到 $\phi$.

### 3.5.1 球域问题

**Lem. 6 球坐标下的 Laplace 方程**

$$\nabla^2\phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\phi}{\partial\varphi^2} = 0 \tag{3.34}$$

**Thm. 6 球域 Laplace 方程的通解**

设电势为 $\phi = \phi(r,\theta,\varphi) = R(r)\Theta(\theta)\Phi(\varphi)$, 可得通解

$$\phi = \sum_{n=0}^{\infty}\sum_{m=0}^{n}\left[\left(A_n^m r^n + \frac{B_n^m}{r^{n+1}}\right)\mathrm{P}_n^m(\cos\theta)\cos m\varphi + \left(C_n^m r^n + \frac{D_n^m}{r^{n+1}}\right)\mathrm{P}_n^m(\cos\theta)\sin m\varphi\right] \tag{3.35}$$

**Thm. 7 轴对称通解**

若 $\dfrac{\partial\phi}{\partial\varphi} = 0$, 取 $m = 0$ 可得轴对称通解

$$\phi = \sum_{n=0}^{\infty}\left(A_n r^n + \frac{B_n}{r^{n+1}}\right)\mathrm{P}_n(\cos\theta) \tag{3.36}$$

**Thm. 8 球对称通解**

若 $\dfrac{\partial\phi}{\partial\theta} = \dfrac{\partial\phi}{\partial\varphi} = 0$, 取 $n = m = 0$ 可得球对称通解

$$\phi = A + \frac{B}{r} \tag{3.37}$$

### 3.5.2 圆柱域问题 *

### 3.5.3 保角变换 *

### 3.5.4 圆域问题

**Lem. 7 极坐标下的 Laplace 方程**

$$\nabla^2\phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2\phi}{\partial\theta^2} = 0 \tag{3.38}$$

**Thm. 9 圆域 Laplace 方程的通解**

设电势为 $\phi(r,\theta) = R(r)\Theta(\theta)$, 可得通解

$$\phi = A_0 + B_0\ln r + \sum_{n=1}^{\infty}(A_n r^n + B_n r^{-n})(C_n\cos n\theta + D_n\sin n\theta) \tag{3.39}$$

## 3.6 电多极矩 *

多级展开条件: 源相对集中于一个小区域, 且线度远远小于源点到场点的距离.

### 3.6.1 静电势的多极展开

$$\phi(\boldsymbol{x}) \tag{3.40}$$

### 3.6.2 电偶极矩

**Prop. 8 电偶极子的电势**

$$\phi_p = \frac{1}{4\pi\varepsilon_0}\frac{\boldsymbol{p}\cdot\boldsymbol{r}}{r^3} \tag{3.41}$$

**Prop. 9 电偶极子的电场**

$$\boldsymbol{E}_p = \frac{1}{4\pi\varepsilon_0}\frac{3(\boldsymbol{p}\cdot\boldsymbol{e}_r)\boldsymbol{e}_r - \boldsymbol{p}}{r^3} \tag{3.42}$$

**Prop. 10 电偶极子的电荷密度**

$$\rho_p = -\boldsymbol{p}\cdot\boldsymbol{\nabla}\delta(\boldsymbol{r}) \tag{3.43}$$

### 3.6.3 电势极矩

$$D_{\alpha\beta} = \int (3x'_\alpha x'_\beta - r'^2 \delta_{\alpha\beta}) \rho(\boldsymbol{x}') \,\mathrm{d}^3 x' \tag{3.44}$$

### 3.6.4 静电能量的多极展开

### 3.6.5 球域中的电多极矩

**Lem. 8 球谐函数的加法公式**

$$\mathrm{P}_l(\cos\gamma) = \sum_{m=-l}^{+l} \frac{(l-m)!}{(l+m)!} \mathrm{P}_l^m(\cos\theta) \mathrm{P}_l^m(\cos\theta') \mathrm{e}^{\mathrm{i}m(\varphi-\varphi')} \tag{3.45}$$

其中 $\cos\gamma = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\varphi-\varphi')$.

$$A_{lm} = \frac{4\pi}{2l+1} \int_{V'} \rho(\boldsymbol{x}') x'^l \mathrm{Y}^*_{lm}(\theta', \varphi') \tag{3.46}$$

$$B_{lm} = \frac{4\pi}{2l+1} \int_{V'} \rho(\boldsymbol{x}') \frac{1}{x'^{l+1}} \mathrm{Y}^*_{lm}(\theta', \varphi') \tag{3.47}$$

# 4 静磁学

**Lem. 1 静磁场基本方程**

$$\begin{cases} \nabla\times\boldsymbol{H}=\boldsymbol{J} \\ \nabla\cdot\boldsymbol{B}=0 \\ \boldsymbol{B}=\mu\boldsymbol{H} \\ \begin{cases} \boldsymbol{n}\cdot(\boldsymbol{B}_1-\boldsymbol{B}_2)=0 \\ \boldsymbol{n}\times(\boldsymbol{H}_1-\boldsymbol{H}_2)=\boldsymbol{K} \end{cases} \end{cases} \tag{4.1}$$

## 4.1 磁矢势

### 4.1.1 磁矢势的微分方程

**Prop. 1 磁感应强度的磁矢势表示**

$$\nabla\times\boldsymbol{A}=\boldsymbol{B} \tag{4.2}$$

**Thm. 1 磁矢势的规范不变性**

磁矢势满足规范不变性

$$\boldsymbol{B}'=\nabla\times\boldsymbol{A}'=\nabla\times(\boldsymbol{A}+\nabla\chi)=\nabla\times\boldsymbol{A}+\underbrace{\nabla\times(\nabla\chi)}_{=0}=\boldsymbol{B} \tag{4.3}$$

一般静磁场采用 Coulomb 规范

$$\nabla\cdot\boldsymbol{A}(\boldsymbol{x})=0 \tag{4.4}$$

**Thm. 2 磁矢势的 Poisson 方程**

$$\nabla^2\boldsymbol{A}=-\mu\boldsymbol{J} \tag{4.5}$$

**Pf** $\nabla\times\boldsymbol{B}=\mu\nabla\times\boldsymbol{H}=\mu\boldsymbol{J}=\nabla\times(\nabla\times\boldsymbol{A})=\nabla\underbrace{(\nabla\cdot\boldsymbol{A})}_{=0}-\nabla^2\boldsymbol{A}=-\mu\boldsymbol{J}.$

**Eg. 1 无限长带电直导线**

无限长带电直导线沿 $z$ 轴通有电流 $I$, 设距离导线 $r_0$ 处为令矢势面, 则 $r$ 处的矢势为

$$\boldsymbol{A}(r)=-\frac{\mu_0 I}{2\pi}\ln\frac{r}{r_0}\boldsymbol{e}_z \tag{4.6}$$

**Eg. 2 均匀磁场**

设 $\boldsymbol{B} = B_0\boldsymbol{e}_z$, 则矢势为 $\forall\alpha \in \mathbb{R}$

$$\boldsymbol{A}(x, y) = [-\alpha y\boldsymbol{e}_x + (1-\alpha)x\boldsymbol{e}_y]B_0 \tag{4.7}$$

### 4.1.2 边值关系

**Thm. 3 磁矢势的边值关系**

$$\begin{cases} A_{2t} = A_{1t} \xRightarrow{\text{Coulomb 规范}} \boldsymbol{A}_2 = \boldsymbol{A}_1 \\ \boldsymbol{n} \times \left(\boldsymbol{\nabla} \times \dfrac{\boldsymbol{A}_2}{\mu_2} - \boldsymbol{\nabla} \times \dfrac{\boldsymbol{A}_1}{\mu_1}\right) = \boldsymbol{K} \end{cases} \tag{4.8}$$

**Pf** $\displaystyle\oint_{\partial\Sigma} \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{x} = \int_{\Sigma} \boldsymbol{B} \cdot \mathrm{d}^2\boldsymbol{x} \to 0, \Rightarrow A_{2t} = A_{1t}$. $\boldsymbol{\nabla} \cdot \boldsymbol{A} = 0, \Rightarrow A_{2n} = A_{1n}, \Rightarrow \boldsymbol{A}_2 = \boldsymbol{A}_1$.
$\displaystyle\boldsymbol{K} = \boldsymbol{n} \times (\boldsymbol{H}_2 - \boldsymbol{H}_1) = \boldsymbol{n} \times \left(\boldsymbol{\nabla} \times \frac{\boldsymbol{A}_2}{\mu_2} - \boldsymbol{\nabla} \times \frac{\boldsymbol{A}_1}{\mu_1}\right)$.

### 4.1.3 静磁场的能量

**Prop. 2 磁场的总能量**

设 $\Omega$ 为电流分布区域, 则

$$W = \frac{1}{2}\int_{\Omega} \boldsymbol{A} \cdot \boldsymbol{J}\, \mathrm{d}^3x \tag{4.9}$$

**Pf** $\displaystyle\boldsymbol{B} \cdot \boldsymbol{H} = \boldsymbol{\nabla} \cdot (\boldsymbol{A} \times \boldsymbol{H}) + \boldsymbol{A} \cdot \boldsymbol{J}, \Rightarrow W_\infty = \frac{1}{2}\int_\infty \boldsymbol{B} \cdot \boldsymbol{H}\, \mathrm{d}^3x = \frac{1}{2}\int_\Omega \boldsymbol{A} \cdot \boldsymbol{J}\, \mathrm{d}^3x$.

**Eg. 3 无限长载流线圈总磁能**

$$W = \frac{1}{2}I\Phi \tag{4.10}$$

**Prop. 3 磁相互作用能**

$$W_{\text{int}} = \int_{\Omega'} \boldsymbol{A}_{\text{ext}} \cdot \boldsymbol{J}\, \mathrm{d}^3x' - W_\varepsilon \tag{4.11}$$

其中 $W_\varepsilon$ 为外电源为了维持电流体系恒定, 而额外做的功.

### 4.1.4 Aharonov-Bohm 效应

**Thm. 4 A-B 效应本质**

粒子运动轨迹所围的电磁通量改变了粒子波函数的相位差.

螺线管未通电时, 两束电子波函数相位差为

$$\varphi_0 \approx k_0 d \sin\theta \tag{4.12}$$

通电后相位差为

$$\varphi_1 = \varphi_0 - \frac{e}{\hbar}\oint_C \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{x} \tag{4.13}$$

相位差的改变量为

$$\Delta\varphi = \varphi_1 - \varphi_0 = -\frac{e}{\hbar}\varPhi \tag{4.14}$$

$n$ 级干涉亮纹移动

$$\Delta x = f\Delta(\tan\theta) \approx \frac{e\varPhi f}{mvd} \tag{4.15}$$

## 4.2 静磁场唯一性定理 *

**Thm. 5 静磁场唯一性定理**

## 4.3 磁标势

**Prop. 4 引入磁标势的条件**

若区域 $\varOmega$ 内不存在自由电流或传导电流体分布, 且其边界面 $\partial\varOmega$ 将 $\varOmega$ 分隔为一个单连通区域, 即任意回路都不被自由电流所链环

$$\begin{cases} \boldsymbol{\nabla}\times\boldsymbol{H} = 0 \\ \displaystyle\oint_{\forall L\subset\varOmega} \boldsymbol{H}\cdot \mathrm{d}\boldsymbol{x} = 0 \end{cases} \tag{4.16}$$

则可在 $\varOmega$ 中定义标量势 $\phi_{\mathrm{m}}$ 使得

$$\boldsymbol{H} = -\boldsymbol{\nabla}\phi_{\mathrm{m}} \tag{4.17}$$

对于非铁磁的线性介质

$$\boldsymbol{B} = -\mu\boldsymbol{\nabla}\phi_{\mathrm{m}} \tag{4.18}$$

**Thm. 6　磁标势的 Poisson 方程**

定义等效磁荷 $\rho_{\mathrm{m}} = -\mu_0 \boldsymbol{\nabla} \cdot \boldsymbol{M}$, 则有

$$\nabla^2 \phi_{\mathrm{m}} = -\frac{\rho_{\mathrm{m}}}{\mu_0} \tag{4.19}$$

**Pf**　$\boldsymbol{\nabla} \cdot \boldsymbol{B} = \mu_0 \boldsymbol{\nabla} \cdot (\boldsymbol{H} + \boldsymbol{M}) = 0, \Rightarrow \boldsymbol{\nabla} \cdot \boldsymbol{H} = -\nabla^2 \phi_{\mathrm{m}} = -\boldsymbol{\nabla} \cdot \boldsymbol{M} = \dfrac{\rho_{\mathrm{m}}}{\mu_0}$

**Thm. 7　磁标势的边值关系**

由 $H_{1t} = H_{2t}, B_{1n} = B_{2n}$ 可得

$$\begin{cases} \phi_1 = \phi_2 \\ \dfrac{\partial \phi_1}{\partial n} - \dfrac{\partial \phi_2}{\partial n} = \boldsymbol{n} \cdot (\boldsymbol{M}_1 - \boldsymbol{M}_2) \end{cases} \tag{4.20}$$

对于非铁磁各向同性介质, 由 $\boldsymbol{B} = \mu \boldsymbol{H}$ 可得

$$\begin{cases} \phi_1 = \phi_2 \\ \mu_1 \dfrac{\partial \phi_1}{\partial n} = \mu_2 \dfrac{\partial \phi_2}{\partial n} \end{cases} \tag{4.21}$$

**静电场与静磁场的对比**

$$\boldsymbol{E} = -\boldsymbol{\nabla} \phi \qquad \boldsymbol{H} = -\boldsymbol{\nabla} \phi_{\mathrm{m}}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \frac{\rho_{\mathrm{f}} + \rho_{\mathrm{P}}}{\varepsilon_0} \qquad \boldsymbol{\nabla} \cdot \boldsymbol{H} = \frac{\rho_{\mathrm{m}}}{\mu_0}$$

$$\rho_{\mathrm{P}} = -\boldsymbol{\nabla} \cdot \boldsymbol{P} \qquad \rho_{\mathrm{m}} = -\mu_0 \boldsymbol{\nabla} \cdot \boldsymbol{M}$$

$$\nabla^2 \phi = -\frac{\rho + \rho_{\mathrm{P}}}{\varepsilon_0} \qquad \nabla^2 \phi_{\mathrm{m}} = -\frac{\rho_{\mathrm{m}}}{\mu_0}$$

$$\begin{cases} \phi_2 = \phi_1 \\ \varepsilon_2 \dfrac{\partial \phi_2}{\partial n} = \varepsilon_1 \dfrac{\partial \phi_1}{\partial n} \end{cases} \qquad \begin{cases} \phi_2 = \phi_1 \\ \mu_2 \dfrac{\partial \phi_2}{\partial n} = \mu_1 \dfrac{\partial \phi_1}{\partial n} \end{cases}$$

## 4.4　磁多极矩 *

**Thm. 8　磁矢势的多极展开**

$$\boldsymbol{A}(\boldsymbol{x}) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{x}')}{r} \, \mathrm{d}^3 x = \boldsymbol{A}^{(0)}(\boldsymbol{x}) + \boldsymbol{A}^{(1)}(\boldsymbol{x}) + \boldsymbol{A}^{(2)}(\boldsymbol{x}) + \cdots \tag{4.22}$$

$$\boldsymbol{A}^{(0)}(\boldsymbol{x}) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{x}')}{x} \, \mathrm{d}^3 x' \tag{4.23}$$

$$\boldsymbol{A}^{(1)}(\boldsymbol{x}) = -\frac{\mu_0}{4\pi}\int \boldsymbol{J}(\boldsymbol{x}')(\boldsymbol{x}'\cdot\boldsymbol{\nabla})\frac{1}{x}\,\mathrm{d}^3x' \tag{4.24}$$

$$\boldsymbol{A}^{(2)}(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\int \boldsymbol{J}(\boldsymbol{x}')\frac{1}{2}\boldsymbol{x}'\boldsymbol{x}' : \boldsymbol{\nabla}\boldsymbol{\nabla}\frac{1}{x}\,\mathrm{d}^3x' \tag{4.25}$$

# 5 电磁波

## 5.1 电磁波基本方程

**Lem. 1 电磁波波动方程**

在各向同性的线性均匀非色散介质中

$$\begin{cases}\left(\nabla^2-\mu\varepsilon\dfrac{\partial^2}{\partial t^2}\right)\boldsymbol{E}(t,\boldsymbol{x})=0\\\left(\nabla^2-\mu\varepsilon\dfrac{\partial^2}{\partial t^2}\right)\boldsymbol{B}(t,\boldsymbol{x})=0\end{cases}\tag{5.1}$$

**Rmk** 可得电磁波的传播速度为 $v=\dfrac{1}{\sqrt{\mu\varepsilon}}=\dfrac{c}{\sqrt{\mu_\mathrm{r}\varepsilon_\mathrm{r}}}=\dfrac{c}{n}$.

### 5.1.1 时谐电磁波

**Def. 1 时谐电磁波**

时谐电磁波的场量为

$$\boldsymbol{E}(t,\boldsymbol{x})=\boldsymbol{E}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}\qquad\boldsymbol{B}(t,\boldsymbol{x})=\boldsymbol{B}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}\tag{5.2}$$

**Thm. 1 Helmholtz 方程**

时谐电磁波的波动方程可由满足 $\boldsymbol{\nabla}\cdot\boldsymbol{E}=0$, $\boldsymbol{\nabla}\cdot\boldsymbol{B}=0$ 的 Helmholtz 方程描述

$$\nabla^2\boldsymbol{E}(\boldsymbol{x})+k^2\boldsymbol{E}(\boldsymbol{x})=0\qquad\nabla^2\boldsymbol{B}(\boldsymbol{x})+k^2\boldsymbol{B}(\boldsymbol{x})=0\tag{5.3}$$

其中 $k=\omega\sqrt{\mu\varepsilon}=\dfrac{\omega}{v}$ 为波数.

**Pf** $\dfrac{\partial}{\partial t}\boldsymbol{E}(t,\boldsymbol{x})=-\mathrm{i}\omega\boldsymbol{E}(\boldsymbol{x})$, $\dfrac{\partial^2}{\partial t^2}\boldsymbol{E}(t,\boldsymbol{x})=-\omega^2\boldsymbol{E}(\boldsymbol{x})$.

**Prop. 1 电场与磁场的关系**

$$\boldsymbol{\nabla}\times\boldsymbol{E}(\boldsymbol{x})=\mathrm{i}\omega\boldsymbol{B}(\boldsymbol{x})\qquad\boldsymbol{E}(\boldsymbol{x})=\mathrm{i}\frac{v}{k}\boldsymbol{\nabla}\times\boldsymbol{B}(\boldsymbol{x})\tag{5.4}$$

**Pf** $\boldsymbol{\nabla}\times\boldsymbol{E}(t,\boldsymbol{x})=\mathrm{e}^{-\mathrm{i}\omega t}\boldsymbol{\nabla}\times\boldsymbol{E}(\boldsymbol{x})=-\dfrac{\partial}{\partial t}\boldsymbol{B}(t,\boldsymbol{x})=\mathrm{i}\omega\boldsymbol{B}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}$.

$\boldsymbol{\nabla}\times\boldsymbol{B}(t,\boldsymbol{x})=\mathrm{e}^{-\mathrm{i}\omega t}\boldsymbol{\nabla}\times\boldsymbol{B}(\boldsymbol{x})=\mu\varepsilon\dfrac{\partial}{\partial t}\boldsymbol{E}(t,\boldsymbol{x})=-\mathrm{i}\omega\mu\varepsilon\boldsymbol{E}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}$, where $\omega\mu\varepsilon=\dfrac{\omega}{v^2}=\dfrac{k}{v}$.

### 5.1.2 平面单色波

**Def. 2 平面单色波**

平面单色波的场量为

$$\boldsymbol{E}(t,\boldsymbol{x})=\boldsymbol{E}_0\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}\qquad \boldsymbol{B}(t,\boldsymbol{x})=\boldsymbol{B}_0\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)} \tag{5.5}$$

平面波的相位为

$$\varphi=\boldsymbol{k}\cdot\boldsymbol{x}-\omega t+\varphi_0 \tag{5.6}$$

其中初相位因子 $\mathrm{e}^{\mathrm{i}\varphi_0}$ 一般并入复振幅 $\boldsymbol{E}_0,\boldsymbol{B}_0$ 中.

**Def. 3 相速度**

同一时刻 $\varphi$ 相同的点构成的称为等相面. 电磁波等相面的运动速度, 称为相速度.

$$\boldsymbol{v}_\mathrm{p}=v\boldsymbol{e}_k=\frac{\omega}{k}\boldsymbol{e}_k \tag{5.7}$$

**Def. 4 群速度**

色散关系描述了频率与波矢的关系

$$k=\pm\omega\sqrt{\mu\varepsilon} \tag{5.8}$$

定义相速度为

$$v_\mathrm{g}=\frac{\mathrm{d}\omega}{\mathrm{d}k} \tag{5.9}$$

**Prop. 2 平面单色波的性质**

- 为 TEM 横电磁模波, 即 $\boldsymbol{E}\perp\boldsymbol{k}$, $\boldsymbol{B}\perp\boldsymbol{k}$, $\boldsymbol{E}\perp\boldsymbol{B}$.
- $\boldsymbol{E}(\boldsymbol{x})$ 与 $\boldsymbol{B}(\boldsymbol{x})$ 满足

$$\boldsymbol{E}(\boldsymbol{x})=v\boldsymbol{B}(\boldsymbol{x})\times\boldsymbol{e}_k \tag{5.10}$$

- $\boldsymbol{E},\boldsymbol{B}$ 同相位, 振幅之比为

$$\frac{E_0}{B_0}=v \tag{5.11}$$

  其中 $E_0=\|\boldsymbol{E}_0\|$, $B_0=\|\boldsymbol{B}_0\|$.

- 群速度等于相速度, 即

$$v_\mathrm{g}=\frac{\mathrm{d}}{\mathrm{d}k}\frac{k}{\sqrt{\mu\varepsilon}}=\frac{1}{\sqrt{\mu\varepsilon}}=v_\mathrm{p} \tag{5.12}$$

**Pf** $\nabla\cdot\boldsymbol{E}(t,\boldsymbol{x})=\boldsymbol{E}_0\cdot\nabla\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}=\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{E}=0$, 同理 $\boldsymbol{k}\cdot\boldsymbol{B}=0$.

$$\boldsymbol{E}(\boldsymbol{x})=\mathrm{i}\frac{v}{k}\boldsymbol{\nabla}\times\left(\boldsymbol{B}_0\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}\right)=\mathrm{i}\frac{v}{k}\underbrace{\boldsymbol{\nabla}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}}_{=\mathrm{i}\boldsymbol{k}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{x}}}\times\boldsymbol{B}_0=v\boldsymbol{B}\times\boldsymbol{e}_k.$$

**Def. 5 电磁波阻抗**

$$Z=\sqrt{\frac{\mu}{\varepsilon}}=Z_0\sqrt{\frac{\mu_\mathrm{r}}{\varepsilon_\mathrm{r}}} \tag{5.13}$$

其中 $Z_0=\sqrt{\mu_0/\varepsilon_0}\approx 377\ \Omega$ 为真空中波阻抗.

**Rmk** $Z\boldsymbol{H}=\boldsymbol{e}_k\times\boldsymbol{E}$, $E_0/H_0=Z$.

### 5.1.3 能量与能流

**Prop. 3 平面单色波的能量密度**

平面电磁波的电场能量与磁场能量相等.

$$\begin{aligned} w(t,\boldsymbol{x}) &= \frac{1}{2}\varepsilon E^2+\frac{1}{2\mu}B^2 \\ &= \varepsilon E^2=\varepsilon E_0^2\cos^2(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t) \\ &= \frac{1}{\mu}B^2=\frac{1}{\mu}B_0^2\cos^2(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t) \end{aligned} \tag{5.14}$$

$$\langle w\rangle=\int_0^T w(t,\boldsymbol{x})\,\mathrm{d}t=\frac{1}{2}\varepsilon E_0^2=\frac{1}{2\mu}B_0^2 \tag{5.15}$$

**Prop. 4 平面单色波的能流密度**

$$\boldsymbol{S}(t,\boldsymbol{x})=\boldsymbol{E}\times\boldsymbol{H}=\frac{1}{Z}E^2\boldsymbol{e}_k=v_\mathrm{p}w\boldsymbol{e}_k \tag{5.16}$$

$$\langle\boldsymbol{S}\rangle=\frac{1}{2Z}E_0^2\boldsymbol{e}_k=v_\mathrm{p}\langle w\rangle\,\boldsymbol{e}_k \tag{5.17}$$

**Def. 6 能量传播速度**

$$\boldsymbol{v}_\mathrm{e}=\frac{\langle\boldsymbol{S}\rangle}{\langle w\rangle} \tag{5.18}$$

**Rmk** 对于平面单色波 $\boldsymbol{v}_\mathrm{e}=\boldsymbol{v}_\mathrm{p}=\boldsymbol{v}_\mathrm{g}$.

## 5.2 偏振

**Def. 7 偏振方向**

由于 $\boldsymbol{E}\perp\boldsymbol{k}$, 取 $\boldsymbol{E}$ 的振荡方向为电磁波的偏振方向.

**Thm. 2 电矢量的分解**

选取标准正交基 $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \boldsymbol{e}_3\}$, 令 $\boldsymbol{e}_k = \boldsymbol{e}_3$, 设相位 $\varphi = \boldsymbol{k} \cdot \boldsymbol{x} - \omega t$, 可将复电矢量分解为

$$\tilde{\boldsymbol{E}}(t, \boldsymbol{x}) = \left(E_{10}\mathrm{e}^{\mathrm{i}\delta_1}\boldsymbol{e}_1 + E_{20}\mathrm{e}^{\mathrm{i}\delta_2}\boldsymbol{e}_2\right)\mathrm{e}^{\mathrm{i}\varphi} \tag{5.19}$$

### 5.2.1 椭圆偏振

**Prop. 5 椭圆偏振方程**

设 $\boldsymbol{E} = E_1\boldsymbol{e}_1 + E_2\boldsymbol{e}_2$, 令 $\delta = \delta_2 - \delta_1$, 有电矢量末端点在 $\boldsymbol{e}_1$-$\boldsymbol{e}_2$ 平面形成的椭圆轨迹方程

$$\left(\frac{E_1}{E_{10}}\right)^2 + \left(\frac{E_2}{E_{20}}\right)^2 - 2\frac{E_1E_2}{E_{10}E_{20}}\cos\delta = \sin^2\delta \tag{5.20}$$

**Pf**

$$\boldsymbol{E} = \mathrm{Re}\{\tilde{\boldsymbol{E}}\} = E_{10}\cos(\delta_1 + \varphi)\boldsymbol{e}_1 + E_{20}\cos(\delta_2 + \varphi)\boldsymbol{e}_2 = E_1\boldsymbol{e}_1 + E_2\boldsymbol{e}_2$$

$$\Longrightarrow \begin{cases} \dfrac{E_1}{E_{10}} = \cos(\delta_1 + \varphi) \equiv X & \Longrightarrow \varphi = \arccos X - \delta_1 \\ \dfrac{E_2}{E_{20}} = \cos(\delta_2 + \varphi) \equiv Y & \Longrightarrow \varphi = \arccos Y - \delta_2 \end{cases}$$

$$\Longrightarrow \arccos Y - \arccos X = \delta \xRightarrow{\sin(\arccos x) = \sqrt{1-x^2}} \begin{cases} \cos\delta = XY + \sqrt{1-X^2}\sqrt{1-Y^2} \\ \sin\delta = X\sqrt{1-Y^2} - Y\sqrt{1-X^2} \end{cases}$$

消去含根号的项: $\sin^2\delta + 2XY\cos\delta = X^2 + Y^2$ 整理可得一般椭圆标准方程

### 5.2.2 线偏振

**Prop. 6 线偏振**

若 $\delta = m\pi$ $(m \in \mathbb{Z})$, 则椭圆偏振变为线偏振

$$\boldsymbol{E} = (E_{10}\boldsymbol{e}_1 \pm E_{20}\boldsymbol{e}_2)\cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega t + \delta_1) \tag{5.21}$$

$m$ 为偶数时取 $+$, 奇数时取 $-$.

### 5.2.3 圆偏振

若 $\delta = \dfrac{\pi}{2} + m\pi$ $(m \in \mathbb{Z})$且 $E_{10} = E_{20} = \dfrac{E_0}{\sqrt{2}}$, 则椭圆偏振变为圆偏振

$$\begin{cases} \boldsymbol{E}_{\mathrm{L}} = \dfrac{E_0}{\sqrt{2}}\left[\cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega t + \delta_1)\boldsymbol{e}_1 + \sin(\boldsymbol{k} \cdot \boldsymbol{x} + \omega t + \delta_1)\boldsymbol{e}_2\right] & \text{左旋圆偏振} \\ \boldsymbol{E}_{\mathrm{R}} = \dfrac{E_0}{\sqrt{2}}\left[\cos(\boldsymbol{k} \cdot \boldsymbol{x} - \omega t + \delta_1)\boldsymbol{e}_1 - \sin(\boldsymbol{k} \cdot \boldsymbol{x} + \omega t + \delta_1)\boldsymbol{e}_2\right] & \text{右旋圆偏振} \end{cases} \tag{5.22}$$

**Def. 8 圆偏振基矢**

$$\boldsymbol{e}_\pm = \frac{1}{\sqrt{2}}(\boldsymbol{e}_1 \pm \mathrm{i}\boldsymbol{e}_2) \tag{5.23}$$

$$\Longrightarrow \begin{cases} \tilde{\boldsymbol{E}}_\mathrm{L} = \dfrac{E_0}{\sqrt{2}}(\boldsymbol{e}_1 + \mathrm{i}\boldsymbol{e}_2)\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)} = \tilde{E}_0\boldsymbol{e}_+ \\ \tilde{\boldsymbol{E}}_\mathrm{R} = \dfrac{E_0}{\sqrt{2}}(\boldsymbol{e}_1 - \mathrm{i}\boldsymbol{e}_2)\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)} = \tilde{E}_0\boldsymbol{e}_- \end{cases} \tag{5.24}$$

$\{\boldsymbol{e}_\pm\}$ 有如下性质

$$\begin{cases} \boldsymbol{e}_\pm \cdot \boldsymbol{e}_3 = 0 & \\ \boldsymbol{e}_\pm \cdot \boldsymbol{e}_\mp^* = \boldsymbol{e}_\mp \cdot \boldsymbol{e}_\pm^* = 0 & \text{正交性} \\ \boldsymbol{e}_\pm \cdot \boldsymbol{e}_\pm^* = 1 & \text{归一性} \end{cases} \tag{5.25}$$

### 5.2.4 Stokes 参数与偏振度

**Def. 9 Stokes 参数**

设 $\boldsymbol{e}_1 \cdot \tilde{\boldsymbol{E}}_0 = a_1\mathrm{e}^{\mathrm{i}\delta_1}$, $\boldsymbol{e}_2 \cdot \tilde{\boldsymbol{E}}_0 = a_2\mathrm{e}^{\mathrm{i}\delta_2}$ $(a_1, a_2 \in \mathbb{R}^+)$, 则 Stokes 参数为

$$s_0 = a_1^2 + a_2^2 \quad s_1 = a_1^2 - a_2^2 \quad s_2 = 2a_1a_2\cos(\delta_2 - \delta_1) \quad s_3 = 2a_1a_2\sin(\delta_2 - \delta_1) \tag{5.26}$$

**Def. 10 偏振度**

偏振度定义为 $(0 \le P \le 1)$:

$$P = \frac{\sqrt{s_1^2 + s_2^2 + s_3^2}}{s_0} \tag{5.27}$$

**Prop. 7**

1. 对于完全偏振光 $(P = 1)$, 满足

$$s_0^2 = s_1^2 + s_2^2 + s_3^2 \tag{5.28}$$

2. 对于部分偏振光, 有 $s_0^2 > s_1^2 + s_2^2 + s_3^2$, $0 < P < 1$.

3. 对于完全非偏振光 (自然光), 则 $s_1 = s_2 = s_3 = 0$, $P = 0$.

## 5.3 电磁波的衍射

### 5.3.1 Kirchhoff 公式

**Thm. 3 Kirchhoff 公式**

$$\psi(\boldsymbol{x}) = -\frac{1}{4\pi}\oint_{\Sigma}\frac{\mathrm{e}^{\mathrm{i}kr}}{r}\left[\boldsymbol{\nabla}'\psi + \left(\mathrm{i}k - \frac{1}{r}\right)\frac{\boldsymbol{r}}{r}\psi\right]\cdot \mathrm{d}^2\boldsymbol{x} \tag{5.29}$$

### 5.3.2 Fraunhofer 衍射

## 5.4 电磁波在介质界面上的反射与折射

### 5.4.1 角度关系

**Thm. 4 反射定律与 Snell 折射定律**

$$\theta = \theta' \tag{5.30}$$

$$n_1 \sin\theta = n_2 \sin\theta'' \tag{5.31}$$

### 5.4.2 Fresnel 公式

**Thm. 5 Fresnel 公式**

$$\left\{\begin{array}{ll} \left(\dfrac{E'}{E}\right)_{\mathrm{p}} = \dfrac{\tan(\theta-\theta'')}{\tan(\theta+\theta'')} & \left(\dfrac{E'}{E}\right)_{\mathrm{s}} = -\dfrac{\sin(\theta-\theta'')}{\sin(\theta+\theta'')} \\ \left(\dfrac{E''}{E}\right)_{\mathrm{p}} = \dfrac{2\cos\theta\sin\theta''}{\sin(\theta+\theta'')\cos(\theta-\theta'')} & \left(\dfrac{E''}{E}\right)_{\mathrm{s}} = \dfrac{2\cos\theta\sin\theta''}{\sin(\theta+\theta'')} \end{array}\right\} \tag{5.32}$$

$$\left\{\begin{array}{ll} \left(\dfrac{E_0'}{E_0}\right)_{\mathrm{p}} = \dfrac{Z_1\cos\theta - Z_2\cos\theta''}{Z_1\cos\theta + Z_2\cos\theta''} & \left(\dfrac{E_0'}{E_0}\right)_{\mathrm{s}} = \dfrac{Z_2\cos\theta - Z_1\cos\theta''}{Z_2\cos\theta + Z_1\cos\theta''} \\ \left(\dfrac{E_0''}{E_0}\right)_{\mathrm{p}} = \dfrac{2Z_2\cos\theta}{Z_1\cos\theta + Z_2\cos\theta''} & \left(\dfrac{E_0''}{E_0}\right)_{\mathrm{s}} = \dfrac{2Z_2\cos\theta}{Z_2\cos\theta + Z_1\cos\theta''} \end{array}\right\} \tag{5.33}$$

**Cor. 1 垂直入射时电场振幅比**

垂直入射时 p 偏振与 s 偏振无区别.

$$\left\{\frac{E_0'}{E_0} = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad \frac{E_0''}{E_0} = \frac{2Z_2}{Z_2 + Z_1}\right\} \tag{5.34}$$

**Prop. 8 阻抗匹配**

### 5.4.3　Brewster 角

**Thm. 6　Brewster 角**

对于 p 偏振, 当入射角 $\theta=\theta_{\mathrm{B}}$ 时, 反射系数为零 $R_{\mathrm{p}}$. 其中 Brewster 角 $\theta_{\mathrm{B}}$ 满足

$$\tan\theta_{\mathrm{B}}=\sqrt{\frac{(Z_1/Z_2)^2-1}{1-(n_1/n_2)^2}}=\sqrt{\frac{\varepsilon_2}{\varepsilon_1}\frac{\varepsilon_2\mu_1-\varepsilon_1\mu_2}{\varepsilon_2\mu_2-\varepsilon_1\mu_1}} \tag{5.35}$$

**Cor. 2　非磁性介质的 Brewster 角**

$$\tan\theta_{\mathrm{B}}=\sqrt{\frac{\varepsilon_2}{\varepsilon_1}}=\frac{n_2}{n_1} \tag{5.36}$$

### 5.4.4　半波损失

### 5.4.5　全反射

**Prop. 9　全反射**

若 $n_1>n_2$, 定义临界角 $\theta_{\mathrm{c}}$ 为

$$\sin\theta_{\mathrm{c}}=\frac{n_2}{n_1} \tag{5.37}$$

当 $\theta=\theta_{\mathrm{c}}$ 时, $\theta''=90^\circ$. 当 $\sin\theta>\dfrac{n_2}{n_1}$ 时, 无法定义实数折射角, 称为全反射现象.

**Prop. 10　消逝波**

$\theta>\theta_{\mathrm{c}}$ 时, 有 $k_1''=k\sin\theta$, $k''=k\sin\theta_{\mathrm{c}}$, 因此

$$k_3''=\mathrm{i}k_1\sqrt{\sin^2\theta-\sin^2\theta_{\mathrm{c}}} \tag{5.38}$$

令 $k_3''=\mathrm{i}\kappa$

$$\kappa=k_1\sqrt{\sin^2\theta-\sin^2\theta_{\mathrm{c}}} \tag{5.39}$$

则折射波的电场为

$$\boldsymbol{E}''=\boldsymbol{E}_0''\mathrm{e}^{-\kappa z}\mathrm{e}^{\mathrm{i}(k_1''x-\omega t)} \tag{5.40}$$

透入介质 2 的厚度与波长同数量级, 约为

$$\kappa^{-1}=\frac{\lambda_1}{2\pi\sqrt{\sin^2\theta-\sin^2\theta_{\mathrm{c}}}} \tag{5.41}$$

其平均能流密度只有 $x$ 分量, 无 $z$ 分量.

$$\frac{E'}{E}=\mathrm{e}^{-2\mathrm{i}\varphi}\qquad\tan\varphi=\frac{\sin^2\theta-\sin^2\theta_{\mathrm{c}}}{\cos\theta} \tag{5.42}$$

### 5.4.6 反射系数与透射系数

**Def. 11 反射系数, 透射系数**

沿界面法线方向反射与折射的能流分量和入射能流分量之比, 称为反射系数 $R$ 与透射系数 $T$.

$$\begin{cases} R = \dfrac{E_0'^2}{E_0^2} \\ T = \dfrac{Z_2^{-1} E_0''^2 \cos\theta''}{Z_1^{-1} E_0^2 \cos\theta} \end{cases} \tag{5.43}$$

**Prop. 11 p 偏振与 s 偏振电磁波**

$$\begin{cases} R_\mathrm{p} = \left(\dfrac{Z_1\cos\theta - Z_2\cos\theta''}{Z_1\cos\theta + Z_2\cos\theta''}\right)^2 & R_\mathrm{s}\left(\dfrac{Z_2\cos\theta - Z_1\cos\theta''}{Z_2\cos\theta + Z_1\cos\theta''}\right)^2 \\ T_\mathrm{p} = \dfrac{4Z_1Z_2\cos\theta\cos\theta''}{(Z_1\cos\theta + Z_2\cos\theta'')^2} & T_\mathrm{s} = \dfrac{4Z_1Z_2\cos\theta\cos\theta''}{(Z_2\cos\theta + Z_1\cos\theta'')^2} \end{cases} \tag{5.44}$$

## 5.5 色散

### 5.5.1 色散介质的本构关系

**Thm. 7 色散的本质**

色散性质源于介质对外场的非瞬时响应. 对于介电常数

$$\boldsymbol{D}(\boldsymbol{x}) = \varepsilon(\omega)\boldsymbol{E}(\boldsymbol{x}) \qquad \varepsilon(\omega) = -\int_{-\infty}^{\infty} \varepsilon(\tau)\mathrm{e}^{\mathrm{i}\omega\tau}\,\mathrm{d}\tau \tag{5.45}$$

**Pf**

$$\begin{cases} \varepsilon(\boldsymbol{x}, \boldsymbol{x}', t, t') \\ \mu^{-1}(\boldsymbol{x}, \boldsymbol{x}', t, t') \end{cases} \xrightarrow[\text{平移不变性}]{\text{空间与时间的}} \begin{cases} \varepsilon(\boldsymbol{x} - \boldsymbol{x}', t - t') \\ \mu^{-1}(\boldsymbol{x} - \boldsymbol{x}', t - t') \end{cases}$$

$$\begin{cases} \boldsymbol{D}(\boldsymbol{x}, t) = \displaystyle\int \varepsilon(\boldsymbol{x} - \boldsymbol{x}', t - t')\boldsymbol{E}(\boldsymbol{x}', t')\,\mathrm{d}^3x'\,\mathrm{d}t' \\ \boldsymbol{H}(\boldsymbol{x}, t) = \displaystyle\int \mu^{-1}(\boldsymbol{x} - \boldsymbol{x}', t - t')\boldsymbol{B}(\boldsymbol{x}', t')\,\mathrm{d}^3x'\,\mathrm{d}t' \end{cases} \quad \underset{\text{因果关系}}{(t > t')}$$

以电场为例, 假设 $\boldsymbol{E}(\boldsymbol{x}, t) = \boldsymbol{E}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}$:

$$\begin{aligned} \boldsymbol{D}(t, \boldsymbol{x}) &= \boldsymbol{E}(\boldsymbol{x}) \int_{-\infty}^{t} \varepsilon(t - t')\mathrm{e}^{-\mathrm{i}\omega t'}\,\mathrm{d}t' = \boldsymbol{E}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t} \int_{-\infty}^{t} \varepsilon(t - t')\mathrm{e}^{\mathrm{i}\omega(t - t')}\,\mathrm{d}t' \\ &\xlongequal{\tau = t - t'} \boldsymbol{E}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t} \int_{0}^{-\infty} \varepsilon(\tau)\mathrm{e}^{\mathrm{i}\omega\tau}\,\mathrm{d}\tau \xlongequal{\varepsilon(\tau) = 0,\ \tau < 0} \varepsilon(\omega)\boldsymbol{E}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t} = \boldsymbol{D}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t} \end{aligned}$$

**Cor. 3**

若磁导率与电导率存在色散, 有线性响应关系

$$\boldsymbol{H}(\boldsymbol{x}) = \mu^{-1}(\omega)\boldsymbol{B}(\boldsymbol{x}) \tag{5.46}$$

$$\boldsymbol{J}(\boldsymbol{x}) = \sigma(\omega)\boldsymbol{E}(\boldsymbol{x}) \tag{5.47}$$

### 5.5.2 Kramers-Kronig 关系

**Thm. 8 Kramers-Kronig 关系**

$$\mathrm{Re}\{\varepsilon_{\mathrm{r}}(\omega)\} = 1 + \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty} \frac{\mathrm{Im}\{\varepsilon_{\mathrm{r}}(\omega')\}}{\omega' - \omega}\,\mathrm{d}\omega' \tag{5.48}$$

$$\mathrm{Im}\{\varepsilon_{\mathrm{r}}(\omega)\} = \frac{1}{\pi}\mathcal{P}\int_{-\infty}^{\infty} \frac{\mathrm{Re}\{\varepsilon_{\mathrm{r}}(\omega') - 1\}}{\omega' - \omega}\,\mathrm{d}\omega' \tag{5.49}$$

### 5.5.3 一维波包在色散介质中的传播

**Def. 12 波包**

波包为一系列单色平面波的线性叠加, 其振幅为

$$\psi(t,x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} A(k)\mathrm{e}^{\mathrm{i}[kx-\omega(k)t]}\,\mathrm{d}k \tag{5.50}$$

其中 $A(k)$ 可由任意时刻 $t_0$ 的振幅作 Fourier 逆变换得到

$$A(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \psi(t_0,x)\mathrm{e}^{-\mathrm{i}\omega(k)t_0}\mathrm{e}^{-\mathrm{i}kx}\,\mathrm{d}x \tag{5.51}$$

**Lem. 2 位置-波数不确定性关系**

$$\Delta k \cdot \Delta x \geq \frac{1}{2} \tag{5.52}$$

**Prop. 12 波包的群速度与相速度**

对于 $n = n(\omega)$ 的色散介质, 相速度为

$$v_{\mathrm{p}} = \frac{\omega}{k} = \frac{c}{n(\omega)} \tag{5.53}$$

群速度为

$$v_{\mathrm{g}} = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{c}{\dfrac{\mathrm{d}}{\mathrm{d}\omega}[\omega n(\omega)]} = \frac{c}{n(\omega) + \omega\dfrac{\mathrm{d}n}{\mathrm{d}\omega}} = \frac{v_{\mathrm{p}}}{1 + \dfrac{\omega}{n}\dfrac{\mathrm{d}n}{\mathrm{d}\omega}} \tag{5.54}$$

因此只要 $\dfrac{\mathrm{d}n}{\mathrm{d}\omega} \neq 0$, 辄 $v_{\mathrm{g}} \neq v_{\mathrm{p}}$.

**Prop. 13 线性色散**

线性色散要求 $A(k)$ 在 $k = k_0$ 附近有一个较小的延展 $\Delta k$, 且 $\omega(k)$ 变化缓慢, 则有

$$\omega(k) \approx \omega(k_0) + \left.\frac{\mathrm{d}\omega}{\mathrm{d}k}\right|_{k=k_0}(k - k_0) = \omega_0 + v_{\mathrm{g}}(k - k_0) \tag{5.55}$$

则波包的时间演化近似为

$$\psi(t, x) = \phi(0, x - v_{\mathrm{g}}t)\mathrm{e}^{\mathrm{i}(k_0 v_{\mathrm{g}} - \omega_0)t} \tag{5.56}$$

线性色散近似下, 波包的包络形状在传播过程中不变, 仅以 $v_{\mathrm{g}}$ 的速率整体平移.

**Pf** $\varphi = kx - \omega(k)t \approx kx - \omega_0 t - v_{\mathrm{g}}kt + v_{\mathrm{g}}k_0 t = k(x - v_{\mathrm{g}}t) + (k_0 v_{\mathrm{g}} - \omega_0)t$.
$\psi(t, x) = \dfrac{1}{\sqrt{2\pi}}\mathrm{e}^{\mathrm{i}(k_0 v_{\mathrm{g}} - \omega_0)t}\displaystyle\int_{-\infty}^{\infty} A(k)\mathrm{e}^{\mathrm{i}k(x - v_{\mathrm{g}}t)}\,\mathrm{d}k = \phi(0, x - v_{\mathrm{g}}t)\mathrm{e}^{\mathrm{i}(k_0 v_{\mathrm{g}} - \omega_0)t}$.

### 5.5.4 非磁性绝缘介质的色散性质

**Prop. 14 平面电磁波在色散介质中的 Maxwell 方程组**

若非磁性的 ($\mu = \mu_0$) 各向同性线性介质的介电常量存在色散, 即 $\varepsilon = \varepsilon(\omega)$, 则平面电磁波在其中的 Maxwell 方程组可写为.

$$\begin{cases} \boldsymbol{k} \times \boldsymbol{E}_0 = \omega \boldsymbol{B}_0 \\ \boldsymbol{k} \times \boldsymbol{B}_0 = -\omega\mu_0\varepsilon(\omega)\boldsymbol{E}_0 \\ \boldsymbol{k} \cdot \boldsymbol{E}_0 = 0 \\ \boldsymbol{k} \cdot \boldsymbol{B}_0 = 0 \end{cases} \tag{5.57}$$

**Pf** 将电磁波场量 $\boldsymbol{E}(t, \boldsymbol{x}) = \boldsymbol{E}_0\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}$, $\boldsymbol{B}(t, \boldsymbol{x}) = \boldsymbol{B}_0\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}$ 代入无源色散介质的 Maxwell 方程可得, 其中磁场旋度式为 $\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0\varepsilon(\omega)\dfrac{\partial \boldsymbol{E}}{\partial t}$.

**Prop. 15 非磁性绝缘介质中的色散关系**

$$k = k_0\sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}} \tag{5.58}$$

其中 $k_0 = \dfrac{\omega}{c}$ 为真空中波矢.

**Pf** 由 $\boldsymbol{k} \cdot \boldsymbol{E}_0 = 0$, 可得 $\boldsymbol{k} \times (\boldsymbol{k} \times \boldsymbol{E}_0) = \omega\boldsymbol{k} \times \boldsymbol{B}_0 = -\underbrace{\omega^2\mu_0\varepsilon(\omega)}_{=k^2}\boldsymbol{E}_0$.

$$\Rightarrow k = \omega\sqrt{\mu_0\varepsilon(\omega)} = \frac{\omega}{c}\sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}} = k_0\sqrt{\frac{\varepsilon(\omega)}{\varepsilon_0}}.$$

### 5.5.5 非磁性导电介质的色散性质

**Prop. 16 导电介质的本构关系**

设导电介质的介电常量存在色散, 则有

$$\boldsymbol{D}(\boldsymbol{x}) = \varepsilon_{\mathrm{b}}(\omega)\boldsymbol{E}(\boldsymbol{x}) \tag{5.59}$$

其中的传导电流满足 Ohm 定律

$$\boldsymbol{J}(\boldsymbol{x}) = \sigma_{\mathrm{c}}\boldsymbol{E}(\boldsymbol{x}) \tag{5.60}$$

**Def. 13 导电介质的等效介电常量**

$$\varepsilon_{\mathrm{eff}} = \varepsilon_{\mathrm{b}}(\omega) + \mathrm{i}\frac{\sigma_{\mathrm{c}}}{\omega} \tag{5.61}$$

**Thm. 9 平面电磁波在导电介质中的 Maxwell 方程组**

$$\begin{cases} \boldsymbol{k}\times\boldsymbol{E}_0 = \omega\boldsymbol{B}_0 \\ \boldsymbol{k}\times\boldsymbol{B}_0 = -\omega\mu_0\left(\varepsilon_{\mathrm{b}} + \mathrm{i}\dfrac{\sigma_{\mathrm{c}}}{\omega}\right)\boldsymbol{E}_0 = -\omega\mu_0\varepsilon_{\mathrm{eff}}\boldsymbol{E}_0 \\ \boldsymbol{k}\cdot\boldsymbol{E}_0 = 0 \\ \boldsymbol{k}\cdot\boldsymbol{B}_0 = 0 \end{cases} \tag{5.62}$$

**Pf** 导电介质中磁场旋度为 $\nabla\times\boldsymbol{B} = \mu_0\boldsymbol{J} + \dfrac{\partial\boldsymbol{D}}{\partial t} = \mu_0\sigma_{\mathrm{c}}\boldsymbol{E} + \mu_0\varepsilon_{\mathrm{b}}(\omega)\dfrac{\partial\boldsymbol{E}}{\partial t}$, 将平面电磁波场量代入得证.

**Prop. 17 非磁性导电介质中的色散关系**

$$k = k_0\sqrt{\frac{\varepsilon_{\mathrm{eff}}}{\varepsilon_0}} \tag{5.63}$$

### 5.5.6 色散介质的穿透深度 (趋肤深度)

**Lem. 3 复介电常量与复波矢量**

色散介质中介电常量与波矢量均为复值, 可显式写出实部与虚部

$$\varepsilon(\omega) = \varepsilon_0(\varepsilon_{\mathrm{r}}' + \mathrm{i}\varepsilon_{\mathrm{r}}'') \tag{5.64}$$

$$\boldsymbol{k}(\omega) = \boldsymbol{\beta}(\omega) + \mathrm{i}\boldsymbol{\alpha}(\omega) \tag{5.65}$$

可得 $\boldsymbol{\alpha}$ 与 $\boldsymbol{\beta}$ 的关系

$$\begin{cases} \beta^2 - \alpha^2 = k_0^2 \varepsilon_{\mathrm{r}}' \\ \boldsymbol{\beta} \cdot \boldsymbol{\alpha} = \dfrac{1}{2} k_0^2 \varepsilon_{\mathrm{r}}'' \end{cases} \tag{5.66}$$

一般 $\boldsymbol{\alpha}$ 与 $\boldsymbol{\beta}$ 方向不同. 若相同, 则可解得

$$\begin{cases} \beta = k_0 \sqrt{\dfrac{\varepsilon_{\mathrm{r}}}{2}} \sqrt{\sqrt{1 + \left(\dfrac{\varepsilon_{\mathrm{r}}''}{\varepsilon_{\mathrm{r}}'}\right)^2} + 1} \\ \alpha = k_0 \sqrt{\dfrac{\varepsilon_{\mathrm{r}}}{2}} \sqrt{\sqrt{1 + \left(\dfrac{\varepsilon_{\mathrm{r}}''}{\varepsilon_{\mathrm{r}}'}\right)^2} - 1} \end{cases} \tag{5.67}$$

**Pf** 方向相同时 $\boldsymbol{\beta} \cdot \boldsymbol{\alpha} = \beta\alpha$, $\Rightarrow \dfrac{k_0^2 \varepsilon_{\mathrm{r}}''}{2\beta}$, $\Rightarrow \beta^4 - k_0^2 \varepsilon_{\mathrm{r}}' \beta^2 - \dfrac{1}{4} k_0^4 \varepsilon_{\mathrm{r}}''^2 = 0$. 利用求根公式可得 $\beta$, 反代得 $\alpha$.

**Prop. 18 色散介质中的电矢量**

$$\boldsymbol{E}(t, \boldsymbol{x}) = \boldsymbol{E}_0 \mathrm{e}^{-\boldsymbol{\alpha} \cdot \boldsymbol{x}} \mathrm{e}^{\mathrm{i}(\boldsymbol{\beta} \cdot \boldsymbol{x} - \omega t)} \tag{5.68}$$

**Rmk** 其中 $\mathrm{e}^{\mathrm{i}(\boldsymbol{\beta} \cdot \boldsymbol{x} - \omega t)}$ 刻画电矢量的振荡, $\mathrm{e}^{-\boldsymbol{\alpha} \cdot \boldsymbol{x}}$ 刻画电矢量的衰减.

**Pf** $\mathrm{e}^{\mathrm{i}(\boldsymbol{k} \cdot \boldsymbol{x} - \omega t)} = \mathrm{e}^{\mathrm{i}[(\boldsymbol{\beta} + \mathrm{i}\boldsymbol{\alpha}) \cdot \boldsymbol{x} - \omega t]} = \mathrm{e}^{-\boldsymbol{\alpha} \cdot \boldsymbol{x} + \mathrm{i}(\boldsymbol{\beta} \cdot \boldsymbol{x} - \omega t)}$.

**Def. 14 穿透深度**

电磁波垂直入射后, 振幅衰减为 $\mathrm{e}^{-1}$ 倍时传播的距离, 称为该介质的穿透深度.

$$\delta = \frac{1}{\alpha} \tag{5.69}$$

**Prop. 19 导电介质中的穿透深度**

对于导电介质, $\varepsilon_{\mathrm{r}}' = \dfrac{\varepsilon_{\mathrm{b}}(\omega)}{\varepsilon_0}$, $\varepsilon_{\mathrm{r}}'' = \dfrac{\sigma_{\mathrm{c}}}{\omega \varepsilon_0}$, 于是有

$$\begin{cases} \beta^2 - \alpha^2 = k_0^2 \dfrac{\varepsilon_{\mathrm{b}}}{\varepsilon_0} = \omega^2 \mu_0 \varepsilon_{\mathrm{b}} \\ \boldsymbol{\beta} \cdot \boldsymbol{\alpha} = \dfrac{1}{2} \omega \mu_0 \sigma_{\mathrm{c}} \end{cases} \tag{5.70}$$

解得

$$\begin{cases} \beta = k_0\sqrt{\dfrac{\varepsilon_\mathrm{b}}{2\varepsilon_0}}\sqrt{\sqrt{1+\left(\dfrac{\sigma_\mathrm{c}}{\omega\varepsilon_\mathrm{b}}\right)^2}+1} \\ \alpha = k_0\sqrt{\dfrac{\varepsilon_\mathrm{b}}{2\varepsilon_0}}\sqrt{\sqrt{1+\left(\dfrac{\sigma_\mathrm{c}}{\omega\varepsilon_\mathrm{b}}\right)^2}-1} \end{cases} \tag{5.71}$$

**Cor. 4 弱导电介质的近似**

弱导电介质的电导率相对较小, 有 $\dfrac{\sigma_\mathrm{c}}{\omega\varepsilon_\mathrm{b}} \ll 1$, 近似得

$$\begin{cases} \beta \approx k_0\sqrt{\dfrac{\varepsilon_\mathrm{b}}{\varepsilon_0}} \\ \alpha \approx \dfrac{\sigma_\mathrm{c}}{2}\sqrt{\dfrac{\mu_0}{\varepsilon_\mathrm{b}}} \end{cases} \tag{5.72}$$

穿透深度为

$$\delta = \frac{2}{\sigma_\mathrm{c}}\sqrt{\frac{\varepsilon_\mathrm{b}}{\mu_0}} \tag{5.73}$$

**Cor. 5 良导电介质的近似**

良导电介质的电导率相对较大, 有 $\dfrac{\sigma_\mathrm{c}}{\omega\varepsilon_\mathrm{b}} \gg 1$, 近似得

$$\alpha \approx \beta \approx k_0\sqrt{\frac{\sigma_\mathrm{c}}{2\omega\varepsilon_0}} = \sqrt{\frac{\omega\mu_0\sigma_\mathrm{c}}{2}} \tag{5.74}$$

穿透深度为

$$\delta = \sqrt{\frac{2}{\omega\mu_0\sigma_\mathrm{c}}} \tag{5.75}$$

### 5.5.7 Lorenz 模型

Lorenz 模型是描述电介质的经典色散理论, 核心是束缚电子的受迫阻尼谐振

$$\varepsilon_\mathrm{r}(\omega) = 1 + \frac{\omega_\mathrm{p}^2}{\omega_0^2 - \omega^2 - \mathrm{i}\gamma\omega} \tag{5.76}$$

$$\mathrm{Re}\{\varepsilon_\mathrm{r}(\omega)\} = 1 + \frac{\omega_\mathrm{p}^2(\omega_0^2-\omega^2)}{(\omega_0^2-\omega^2)+\gamma^2\omega^2} \qquad \mathrm{Im}\{\varepsilon_\mathrm{r}(\omega)\} = \frac{\omega_\mathrm{p}^2\gamma\omega}{(\omega_0^2-\omega^2)+\gamma^2\omega^2} \tag{5.77}$$

其中 $\omega_\mathrm{p}$ 为介质的等离子体频率, $\omega_0$ 为谐振本征频率, $\gamma$ 为阻尼系数.

设束缚电子所受阻尼 $\boldsymbol{F}_{\text{damp}} = -m\gamma\dot{\boldsymbol{x}}$, 受电场力 $\boldsymbol{F}_{\text{e}} = -e\boldsymbol{E}_0\mathrm{e}^{-\mathrm{i}\omega t}$, 运动方程为

$$m\ddot{\boldsymbol{x}} + m\omega_0^2\boldsymbol{x} = -m\gamma\dot{\boldsymbol{x}} - e\boldsymbol{E}_0\mathrm{e}^{-\mathrm{i}\omega t} \xRightarrow{\boldsymbol{x}(t)=\boldsymbol{x}_0\mathrm{e}^{-\mathrm{i}\omega t}} \boldsymbol{x} = \frac{-e\boldsymbol{E}}{m(\omega_0^2 - \omega^2 - \mathrm{i}\gamma\omega)}$$

$$\boldsymbol{P} = -n_{\text{e}}e\boldsymbol{x} = \frac{n_{\text{e}}e^2}{m}\frac{\boldsymbol{E}}{\omega_0^2 - \omega^2 - \mathrm{i}\gamma\omega} \xRightarrow{\boldsymbol{P}=\varepsilon_0\chi\boldsymbol{E}} \chi(\omega) = \frac{\omega_{\text{p}}^2}{\omega_0^2 - \omega^2 - \mathrm{i}\gamma\omega}$$

$$\varepsilon_{\text{r}}(\omega) = 1 + \chi(\omega) = 1 + \frac{\omega_{\text{p}}^2}{\omega_0^2 - \omega^2 - \mathrm{i}\gamma\omega} = 1 + \frac{\omega_{\text{p}}^2(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2) + \gamma^2\omega^2} + \mathrm{i}\frac{\omega_{\text{p}}^2\gamma\omega}{(\omega_0^2 - \omega^2) + \gamma^2\omega^2}$$

### 5.5.8 Drude 模型

Drude 模型是描述金属导体的经典色散理论, 核心是带电粒子气模型近似. 设弛豫时间为 $\tau$, 电导率为

$$\sigma(\omega) = \frac{\varepsilon_0\omega_{\text{p}}^2}{\tau^{-1} - \mathrm{i}\omega} \xrightarrow[\text{长波近似}]{\omega\to 0} \sigma_{\text{c}} = \frac{n_{\text{e}}e^2}{m}\tau \tag{5.78}$$

等效相对介电常数为

$$\varepsilon_{\text{r}}(\omega) = 1 - \frac{\omega_{\text{p}}^2}{\omega^2 + \mathrm{i}\omega\tau^{-1}} \tag{5.79}$$

$$\mathrm{Re}\{\varepsilon_{\text{r}}\} = 1 - \frac{\omega_{\text{p}}^2\tau^2}{1 + \omega^2\tau^2} \qquad \mathrm{Im}\{\varepsilon_{\text{r}}\} = \frac{\omega_{\text{p}}^2\tau}{\omega(1 + \omega^2\tau^2)} \tag{5.80}$$

设自由电子受散射力 $\boldsymbol{F}_{\text{sca}} = -\dfrac{m\boldsymbol{v}}{\tau}$, 受电场力 $\boldsymbol{F}_{\text{e}} = -e\boldsymbol{E}_0\mathrm{e}^{-\mathrm{i}\omega t}$ , 运动方程为

$$m\dot{\boldsymbol{v}} = -m\tau^{-1}\boldsymbol{v} - e\boldsymbol{E}_0\mathrm{e}^{-\mathrm{i}\omega t} \xRightarrow{\boldsymbol{v}(t)=\boldsymbol{v}_0\mathrm{e}^{-\mathrm{i}\omega t}} \boldsymbol{v} = \frac{-e\boldsymbol{E}}{m(\tau^{-1} - \mathrm{i}\omega)}$$

$$\boldsymbol{J}_{\text{f}} = \rho\boldsymbol{v} = \frac{n_{\text{e}}e^2}{m}\frac{\boldsymbol{E}}{\tau^{-1} - \mathrm{i}\omega} \xRightarrow{\boldsymbol{J}_{\text{f}}=\sigma\boldsymbol{E}} \sigma(\omega) = \frac{n_{\text{e}}e^2}{m}\frac{1}{\tau^{-1} - \mathrm{i}\omega} = \frac{\varepsilon_0\omega_{\text{p}}^2}{\tau^{-1} - \mathrm{i}\omega}$$

$$\boldsymbol{\nabla} \times \boldsymbol{H} = \boldsymbol{J}_{\text{f}} + \varepsilon_0\frac{\partial\boldsymbol{E}}{\partial t} = (\sigma(\omega) - \mathrm{i}\omega\varepsilon_0)\boldsymbol{E} = -\mathrm{i}\omega\varepsilon_{\text{eff}}(\omega)\boldsymbol{E} \Longrightarrow \varepsilon_{\text{eff}} = \varepsilon_0 + \frac{\mathrm{i}\sigma}{\omega}$$

## 5.6 电磁波在受限空间中的传播

### 5.6.1 波导的导模特征

**Lem. 4 理想导体的边值关系**

理想导体外表面满足

$$\begin{cases} E_t = B_n = 0 \\ \boldsymbol{n} \times \boldsymbol{H} = \boldsymbol{K} \\ \boldsymbol{n} \cdot \boldsymbol{D} = \sigma \end{cases} \tag{5.81}$$

在导体表面, 电场线与界面正交, 磁感线与界面相切.

**Prop. 20**

设波导管轴线方向为 $z$ 方向, 则导模可表为

$$\begin{cases} \boldsymbol{E}(t,\boldsymbol{x}) = \boldsymbol{E}(x,y)\mathrm{e}^{\mathrm{i}(k_3 z-\omega t)} \\ \boldsymbol{H}(t,\boldsymbol{x}) = \boldsymbol{H}(x,y)\mathrm{e}^{\mathrm{i}(k_3 z-\omega t)} \end{cases} \tag{5.82}$$

$$\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+k^2-k_3^2\right)\begin{Bmatrix}\boldsymbol{E}(x,y)\\ \boldsymbol{H}(x,y)\end{Bmatrix}=0 \tag{5.83}$$

**Thm. 10 导模的横纵分解定理**

(Transverse-Longitudinal Decomposition Theorem for Guided Modes)

电磁场的所有横向分量均可由纵向分量表示

$$\begin{cases} E_x = \dfrac{\mathrm{i}}{k_t^2}\left(k_3\dfrac{\partial E_z}{\partial x}+\omega\mu\dfrac{\partial H_z}{\partial y}\right) \\ E_y = \dfrac{\mathrm{i}}{k_t^2}\left(k_3\dfrac{\partial E_z}{\partial y}-\omega\mu\dfrac{\partial H_z}{\partial x}\right) \end{cases} \tag{5.84}$$

$$\begin{cases} H_x = \dfrac{\mathrm{i}}{k_t^2}\left(-\omega\varepsilon\dfrac{\partial E_z}{\partial y}+k_3\dfrac{\partial H_z}{\partial x}\right) \\ H_y = \dfrac{\mathrm{i}}{k_t^2}\left(\omega\varepsilon\dfrac{\partial E_z}{\partial x}+k_3\dfrac{\partial H_z}{\partial y}\right) \end{cases} \tag{5.85}$$

**Pf** $\dfrac{\partial}{\partial z}\to \mathrm{i}k_3$, $\dfrac{\partial}{\partial t}\to -\mathrm{i}\omega$. $\boldsymbol{\nabla}\times\boldsymbol{E}=-\mu\dfrac{\partial \boldsymbol{H}}{\partial t}=\mathrm{i}\omega\mu\boldsymbol{H}$, $\boldsymbol{\nabla}\times\boldsymbol{H}=\varepsilon\dfrac{\partial \boldsymbol{E}}{\partial t}=-\mathrm{i}\omega\varepsilon\boldsymbol{E}$.

$$\begin{cases} \mathrm{i}k_3E_y+\mathrm{i}\omega\mu H_x=\dfrac{\partial E_z}{\partial y} \\ \mathrm{i}k_3E_x-\mathrm{i}\omega\mu H_y=\dfrac{\partial E_z}{\partial x} \\ \dfrac{\partial E_y}{\partial x}-\dfrac{\partial E_x}{\partial y}=\mathrm{i}\omega\mu H_z \end{cases}\qquad \begin{cases} \mathrm{i}\omega\varepsilon E_x-\mathrm{i}k_3H_y=-\dfrac{\partial H_z}{\partial y} \\ \mathrm{i}\omega\varepsilon E_y+\mathrm{i}k_3H_x=\dfrac{\partial H_z}{\partial x} \\ \dfrac{\partial H_y}{\partial x}-\dfrac{\partial H_x}{\partial y}=-\mathrm{i}\omega\varepsilon E_z \end{cases}$$

联立求解可得证.

**Cor. 6**

单边空心波导管不支持 TEM 模.

### 5.6.2 矩形波导管

**Meth. 1 矩形波导管问题的求解**

以 $u(t,x,y,z)=u(x,y)\mathrm{e}^{\mathrm{i}(k_3z-\omega t)}$ 表示电磁场的任一直角坐标分量, 易得通解

$$u(x,y)=(C_1\cos k_1x+D_1\sin k_1x)(C_2\cos k_2y+D_2\sin k_2y) \tag{5.86}$$

结合 $x,y=0$ 处边界条件, 可得

$$\begin{cases} E_x=A_1\cos k_1x\sin k_2y\,\mathrm{e}^{\mathrm{i}k_3x} \\ E_y=A_2\sin k_1x\cos k_2y\,\mathrm{e}^{\mathrm{i}k_3x} \\ E_z=A_3\sin k_1x\sin k_2y\,\mathrm{e}^{\mathrm{i}k_3x} \end{cases} \tag{5.87}$$

结合 $x=l_1$, $y=l_2$ 处边界条件, 可得

$$k_1=\frac{n_1\pi}{l_1}\quad k_2=\frac{n_2\pi}{l_2}\quad (n_1,n_2=0,1,2,\cdots) \tag{5.88}$$

其中 $n_1,n_2$ 不同时为零. 结合 $\boldsymbol{\nabla}\cdot\boldsymbol{E}=0$, 可得

$$k_1A_1+k_2A_2-\mathrm{i}k_3A_3=0 \tag{5.89}$$

再由 $\boldsymbol{H}=-\dfrac{\mathrm{i}}{\omega\mu}\boldsymbol{\nabla}\times\boldsymbol{E}$ 可得 $H_{x,y,z}$ 的取值

$$\begin{cases} H_x=-\dfrac{1}{\omega\mu}(k_3A_2+\mathrm{i}k_2A_3)\sin k_1x\cos k_2y\,\mathrm{e}^{\mathrm{i}(k_3z-\omega t)} \\ H_y=\dfrac{1}{\omega\mu}(k_3A_1+\mathrm{i}k_1A_3)\cos k_1x\sin k_2y\,\mathrm{e}^{\mathrm{i}(k_3z-\omega t)} \\ H_z=-\dfrac{\mathrm{i}}{\omega\mu}(k_1A_2-k_2A_1)\cos k_1x\cos k_2y\,\mathrm{e}^{\mathrm{i}(k_3z-\omega t)} \end{cases} \tag{5.90}$$

**Prop. 21 截止频率**

要求 $k_3=\sqrt{k^2-k_1^2-k_2^2}$ 为实值, 则有

$$k>k_{\mathrm{cut}}=\pi\sqrt{\left(\frac{n_1}{l_1}\right)^2+\left(\frac{n_2}{l_2}\right)^2} \tag{5.91}$$

截止角频率, 截止角频率, 截止波长分别为

$$\omega_{\mathrm{cut}}=\frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{n_1}{l_1}\right)^2+\left(\frac{n_2}{l_2}\right)^2} \tag{5.92}$$

$$f_{\text{cut}} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{n_1}{l_1}\right)^2 + \left(\frac{n_2}{l_2}\right)^2} \tag{5.93}$$

$$\lambda_{\text{cut}} = \frac{2}{\sqrt{\left(\frac{n_1}{l_1}\right)^2 + \left(\frac{n_2}{l_2}\right)^2}} \tag{5.94}$$

若 $l_1 > l_2$, 则 $\mathrm{TE}_{10}$ 波有最低截止频率与截止波长

$$f_{\text{cut}} = \frac{v}{2l_1} \qquad \lambda_{\text{cut}} = 2l_1 \tag{5.95}$$

### 5.6.3　谐振腔

**Prop. 22　矩形谐振腔**

设 $x = 0, l_1$, $y = 0, l_2$, $z = 0, l_3$ 为金属内腔壁, 腔内为各向同性的均匀线性介质, 介电常量为 $\varepsilon$, 磁导率为 $\mu$. $u(x, y, z)$ 表示 $\boldsymbol{E}, \boldsymbol{H}$ 的任一 Cartesius 坐标分量, 满足 $\nabla^2 u + k^2 u = 0$, 其通解为

$$u(x, y, z) = (C_1 \cos k_1 x + D_1 \sin k_1 x)(C_2 \cos k_2 y + D_2 \sin k_2 y)(C_3 \cos k_3 z + D_3 \sin k_3 z) \tag{5.96}$$

结合边界条件与 $\boldsymbol{\nabla} \cdot \boldsymbol{E} = 0$, 可得其特解为

$$\begin{cases} E_x = A_1 \cos k_1 x \sin k_2 y \sin k_3 z \\ E_y = A_2 \sin k_1 x \cos k_2 y \sin k_3 z \\ E_z = A_3 \sin k_1 x \sin k_2 y \cos k_3 z \end{cases} \tag{5.97}$$

$$k_1 = \frac{n_1 \pi}{l_1} \qquad k_2 = \frac{n_2 \pi}{l_2} \qquad k_3 = \frac{n_3 \pi}{l_3} \qquad (n_1, n_2, n_3 = 0, 1, 2, \cdots) \tag{5.98}$$

$$k_1 A_1 + k_2 A_2 + k_3 A_3 = 0 \tag{5.99}$$

谐振腔的共振频率为

$$\omega = \frac{\pi}{\sqrt{\mu\varepsilon}}\sqrt{\frac{n_1^2}{l_1^2} + \frac{n_2^2}{l_2^2} + \frac{n_3^2}{l_3^2}} \tag{5.100}$$

## 5.7 电磁辐射

### 5.7.1 电磁场的势

**Def. 15 标势**

$$-\boldsymbol{\nabla}\phi(t,\boldsymbol{x}) = \boldsymbol{E}(t,\boldsymbol{x}) + \frac{\partial \boldsymbol{A}(t,\boldsymbol{x})}{\partial t} \tag{5.101}$$

**Rmk** 时变电场的势表述为

$$\boldsymbol{E}(t,\boldsymbol{x}) = -\boldsymbol{\nabla}\phi(t,\boldsymbol{x}) - \frac{\partial \boldsymbol{A}(t,\boldsymbol{x})}{\partial t} \tag{5.102}$$

**Pf** 由 $\boldsymbol{\nabla}\cdot\boldsymbol{B}(t,\boldsymbol{x})$, 引入矢势 $\boldsymbol{B}(t,\boldsymbol{x}) = \boldsymbol{\nabla}\times\boldsymbol{A}(t,\boldsymbol{x})$. 于是 $\boldsymbol{\nabla}\times\boldsymbol{E}(t,\boldsymbol{x}) = -\dfrac{\partial}{\partial t}\boldsymbol{\nabla}\times\boldsymbol{A}(t,\boldsymbol{x})$, 从而 $\boldsymbol{\nabla}\times\underbrace{\left[\boldsymbol{E}(t,\boldsymbol{x}) + \dfrac{\partial \boldsymbol{A}(t,\boldsymbol{x})}{\partial t}\right]}_{=-\boldsymbol{\nabla}\phi(t,\boldsymbol{x})} = 0$.

**Lem. 5 有源电磁场的波动方程**

$$\begin{cases} \Box \boldsymbol{E} = \dfrac{1}{\varepsilon_0}\boldsymbol{\nabla}\rho + \mu_0\dfrac{\partial \boldsymbol{J}}{\partial t} \\ \Box \boldsymbol{B} = -\mu_0\boldsymbol{\nabla}\times\boldsymbol{J} \end{cases} \tag{5.103}$$

**Prop. 23 电磁势的 Maxwell 方程组**

$$\begin{cases} \nabla^2\phi + \dfrac{\partial}{\partial t}\boldsymbol{\nabla}\cdot\boldsymbol{A} = -\dfrac{\rho}{\varepsilon_0} \\ \Box \boldsymbol{A} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \dfrac{1}{c^2}\dfrac{\partial\phi}{\partial t}\right) = -\mu_0\boldsymbol{J} \end{cases} \tag{5.104}$$

**Pf** $\boldsymbol{\nabla}\cdot\boldsymbol{E} = \boldsymbol{\nabla}\cdot\left(-\boldsymbol{\nabla}\phi - \dfrac{\partial\boldsymbol{A}}{\partial t}\right) = -\nabla^2\phi - \dfrac{\partial}{\partial t}\boldsymbol{\nabla}\cdot\boldsymbol{A} = \dfrac{\rho}{\varepsilon_0}$, $\Rightarrow \nabla^2\phi + \dfrac{\partial}{\partial t}\boldsymbol{\nabla}\cdot\boldsymbol{A} = -\dfrac{\rho}{\varepsilon_0}$.

$\boldsymbol{\nabla}\times\boldsymbol{B} = \boldsymbol{\nabla}(\boldsymbol{\nabla}\cdot\boldsymbol{A}) - \nabla^2\boldsymbol{A} = \mu_0\boldsymbol{J} + \dfrac{1}{c^2}\dfrac{\partial}{\partial t}\left(-\boldsymbol{\nabla}\phi - \dfrac{\partial\boldsymbol{A}}{\partial t}\right) = \mu_0\boldsymbol{J} - \dfrac{1}{c^2}\boldsymbol{\nabla}\dfrac{\partial\phi}{\partial t} - \dfrac{1}{c^2}\dfrac{\partial^2\boldsymbol{A}}{\partial t^2}$,

$\Rightarrow \nabla^2\boldsymbol{A} - \dfrac{1}{c^2}\dfrac{\partial^2\boldsymbol{A}}{\partial t^2} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \dfrac{1}{c^2}\dfrac{\partial\phi}{\partial t}\right) = -\mu_0\boldsymbol{J}$.

**Prop. 24 Lorentz 力的电磁势表述**

$$\frac{\mathrm{d}\boldsymbol{p}_\mathrm{c}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{p} + q\boldsymbol{A}) = -q\boldsymbol{\nabla}(\phi - \boldsymbol{v}\cdot\boldsymbol{A}) \tag{5.105}$$

**Prop. 25 规范变换**

$$\begin{cases} \boldsymbol{A} & \to \boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\nabla}\psi \\ \phi & \to \phi' = \phi - \dfrac{\partial \psi}{\partial t} \end{cases} \tag{5.106}$$

**Prop. 26**

在 Coulomb 规范 $\boldsymbol{\nabla} \cdot \boldsymbol{A}_{\mathrm{C}}(t, \boldsymbol{x}) = 0$ 下, 方程化为

$$\begin{cases} \nabla^2 \phi_{\mathrm{C}}(t, \boldsymbol{x}) = -\dfrac{\rho(t, \boldsymbol{x})}{\varepsilon_0} \\ \nabla^2 \boldsymbol{A}_{\mathrm{C}}(t, \boldsymbol{x}) - \dfrac{1}{c^2} \dfrac{\partial^2 \boldsymbol{A}_{\mathrm{C}}(t, \boldsymbol{x})}{\partial t^2} - \dfrac{1}{c^2} \dfrac{\partial}{\partial t} \boldsymbol{\nabla} \phi_{\mathrm{C}}(t, \boldsymbol{x}) = -\mu_0 \boldsymbol{J}(t, \boldsymbol{x}) \end{cases} \tag{5.107}$$

其中标势表示为

$$\phi_{\mathrm{C}}(t, \boldsymbol{x}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(t, \boldsymbol{x})}{r} \, \mathrm{d}^3 x \tag{5.108}$$

### 5.7.2 Lorenz 规范与 d'Alembert 方程

**Def. 16 Lorenz 规范**

$$\boldsymbol{\nabla} \cdot \boldsymbol{A}_{\mathrm{L}}(t, \boldsymbol{x}) + \frac{1}{c^2} \frac{\partial \phi_{\mathrm{L}}(t, \boldsymbol{x})}{\partial t} = 0 \tag{5.109}$$

**Prop. 27 d'Alembert 方程**

$$\begin{cases} \Box \phi_{\mathrm{L}}(t, \boldsymbol{x}) = -\dfrac{\rho(t, \boldsymbol{x})}{\varepsilon_0} \\ \Box \boldsymbol{A}_{\mathrm{L}}(t, \boldsymbol{x}) = -\mu_0 \boldsymbol{J}(t, \boldsymbol{x}) \end{cases} \tag{5.110}$$

**Rmk** 形式上, 电荷产生标势波动, 电流产生矢势波动.

**Pf** $\nabla^2 \phi_{\mathrm{L}} + \dfrac{\partial}{\partial t} \boldsymbol{\nabla} \cdot \boldsymbol{A}_{\mathrm{L}} = \nabla^2 \phi_{\mathrm{L}} - \dfrac{1}{c^2} \dfrac{\partial^2 \phi_{\mathrm{L}}}{\partial t^2} = -\dfrac{\rho}{\varepsilon_0}$. $\Box \boldsymbol{A} - \boldsymbol{\nabla} \underbrace{\left( \boldsymbol{\nabla} \cdot \boldsymbol{A} + \dfrac{1}{c^2} \dfrac{\partial \phi}{\partial t} \right)}_{=0} = -\mu_0 \boldsymbol{J}$.

### 5.7.3 推迟势

**Lem. 6 三维无界波动问题的球对称通解**

对于 $\Box \phi = 0$, 若 $\rho(t, \boldsymbol{x}) = \phi(t, r)$, 则方程可化为

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial \phi}{\partial r} \right) - \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = 0 \quad \Longrightarrow \frac{\partial^2 (r\phi)}{\partial r^2} - \frac{1}{c^2} \frac{\partial^2 (r\phi)}{\partial t^2} = 0 \tag{5.111}$$

对于一维波动方程, 有行波解

$$u(t,r)=f_1(t-r/c)+f_2(t+r/c) \tag{5.112}$$

从而球面波解为

$$\phi(t,r)=\frac{1}{r}f_1(t-r/c)+\frac{1}{r}f_2(t+r/c) \tag{5.113}$$

**Def. 17 推迟时刻**

$t$ 时刻的势由 $t-r/c$ 时刻的源决定, 定义此时为推迟时刻

$$t_{\text{rev}}=t-\frac{r}{c} \tag{5.114}$$

$t-t_{\text{rev}}$ 即为电磁相互作用从源点传播到场点所需的时间.

**Prop. 28 原点处含时点电荷的推迟势**

设原点处有一电荷量 $Q(t)$ 随时间变化的点电荷, 其产生的标势 $\phi(t,\boldsymbol{x})$ 的波动方程为

$$\Box\phi(t,\boldsymbol{x})=-\frac{1}{\varepsilon_0}Q(t)\delta(\boldsymbol{x}) \tag{5.115}$$

$r\neq 0$ 时化为齐次波动方程, 解为

$$\phi(t,r)=\frac{Q(t_{\text{rev}})}{4\pi\varepsilon_0 r} \tag{5.116}$$

**Pf** $\boldsymbol{\nabla}t_{\text{rev}}=-\frac{1}{c}\boldsymbol{\nabla}r=-\frac{1}{c}\boldsymbol{e}_r$.

$$\begin{aligned}\nabla^2\frac{Q(t_{\text{rev}})}{r}&=Q(t_{\text{rev}})\nabla^2\frac{1}{r}+2\boldsymbol{\nabla}\frac{1}{r}\cdot\boldsymbol{\nabla}Q(t_{\text{rev}})+\frac{1}{r}\nabla^2Q(t_{\text{rev}})\\&=-4\pi\delta(\boldsymbol{x})Q(t)+2\left(-\frac{1}{r^2}\right)\left(-\frac{1}{c}\frac{\partial Q}{\partial t}\right)+\frac{1}{r}\left(\frac{1}{c^2}\frac{\partial^2Q}{\partial t^2}-\frac{2}{cr}\frac{\partial Q}{\partial t}\right)\\&=\frac{1}{c^2r}\frac{\partial^2Q}{\partial t^2}-4\pi Q(t)\delta(\boldsymbol{x})=\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\frac{Q}{r}-4\pi Q(t)\delta(\boldsymbol{x})\end{aligned}$$

两端除以 $4\pi\varepsilon_0$ 可得 $\Box\phi(t,\boldsymbol{x})=-Q(t)\delta(\boldsymbol{x})/\varepsilon_0$.

**Cor. 7 含时点电荷的标势**

$\boldsymbol{x}'$ 处的点电荷在空间产生的标势为

$$\phi(t,\boldsymbol{x})=\frac{Q(t_{\text{rev}},\boldsymbol{x}')}{4\pi\varepsilon_0 r} \tag{5.117}$$

其中 $r=|\boldsymbol{x}-\boldsymbol{x}'|$.

**Cor. 8 连续带电体的标势**

根据叠加原理, 连续电荷分布 $\rho(t, \boldsymbol{x}')$ 在空间产生的标势为

$$\phi(t, \boldsymbol{x}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(t_{\text{rev}}, \boldsymbol{x}')}{r} \, \mathrm{d}^3 x' \tag{5.118}$$

**Cor. 9 电流的矢势**

含时电流分布 $\boldsymbol{J}(t, \boldsymbol{x}')$ 在空间产生的矢势为

$$\boldsymbol{A}(t, \boldsymbol{x}) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(t_{\text{rev}}, \boldsymbol{x}')}{r} \, \mathrm{d}^3 x' \tag{5.119}$$

**Rmk** 易证, 上述 $\phi(t, \boldsymbol{x})$ 与 $\boldsymbol{A}(t, \boldsymbol{x})$ 满足 Lorenz 规范.

### 5.7.4 时谐电流的多级辐射

**Lem. 7 时谐辐射电磁场**

已知辐射矢势分布 $\boldsymbol{A}(t, \boldsymbol{x})$, 即可求出辐射电磁场.

$$\boldsymbol{B}(t, \boldsymbol{x}) = \boldsymbol{\nabla} \times \boldsymbol{A}(t, \boldsymbol{x}) \tag{5.120}$$

$$\boldsymbol{E}(t, \boldsymbol{x}) = \frac{\mathrm{i}c^2}{\omega} \boldsymbol{\nabla} \times \boldsymbol{B}(t, \boldsymbol{x}) \tag{5.121}$$

**Prop. 29 时谐电流的推迟势**

给定电流分布 $\boldsymbol{J}(t, \boldsymbol{x}') = \boldsymbol{J}(\boldsymbol{x}')\mathrm{e}^{-\mathrm{i}\omega t}$, 可得矢势

$$\boldsymbol{A}(t, \boldsymbol{x}) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{x}')}{r} \mathrm{e}^{\mathrm{i}(kr - \omega t)} \, \mathrm{d}^3 x' = \boldsymbol{A}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t} \tag{5.122}$$

$$\boldsymbol{A}(\boldsymbol{x}) = \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{x}')}{r} \mathrm{e}^{\mathrm{i}kr} \, \mathrm{d}^3 x' \tag{5.123}$$

由 Lorenz 规范可得标势

$$\phi(t, \boldsymbol{x}) = \phi(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t} = \frac{c^2}{\mathrm{i}\omega} \boldsymbol{\nabla} \cdot \boldsymbol{A}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t} \tag{5.124}$$

**Rmk** 也可由电荷守恒定律求出电荷分布

$$\rho(t, \boldsymbol{x}') = \frac{1}{\mathrm{i}\omega} \boldsymbol{\nabla}' \cdot \boldsymbol{J}(t, \boldsymbol{x}') \tag{5.125}$$

再利用连续带电体的推迟势公式求出 $\phi(t, \boldsymbol{x})$.

**Prop. 30 小区域电流产生的辐射**

小区域分布要求 $l \ll r$, $l \ll \lambda$, 其中 $l$ 为电流分布区域的线度, $r = |\boldsymbol{x} - \boldsymbol{x}'|$, $\lambda$ 为辐射波长.

1. $r \ll \lambda$ 为近场区, $\mathrm{e}^{\mathrm{i}kr} \sim 1$, $\boldsymbol{A}(t, \boldsymbol{x}) = \dfrac{\mu_0}{4\pi} \displaystyle\int \frac{\boldsymbol{J}(\boldsymbol{x}')}{r} \, \mathrm{d}^3 x'$, 电磁场的空间依赖形式与静场相似.
2. $r \sim \lambda$ 为感应区.
3. $r \gg \lambda$ 为远场区.

**Rmk** 已知远场区矢势 $\boldsymbol{A}(t, \boldsymbol{x})$, 则电磁场为

$$\boldsymbol{B}(t, \boldsymbol{x}) = \boldsymbol{\nabla} \times \boldsymbol{A}(t, \boldsymbol{x}) = \mathrm{i}k\boldsymbol{e}_R \times \boldsymbol{A}(t, \boldsymbol{x}) \tag{5.126}$$

$$\boldsymbol{E}(t, \boldsymbol{x}) = c\boldsymbol{B}(t, \boldsymbol{x}) \times \boldsymbol{e}_R \tag{5.127}$$

其中 $R = |\boldsymbol{x}|$, $\boldsymbol{e}_R = \dfrac{\boldsymbol{x}}{R}$.

**Prop. 31 时谐电流推迟势的多极展开**

令 $R = |\boldsymbol{x}|$, 在 $r = |\boldsymbol{x} - \boldsymbol{x}'| \gg |\boldsymbol{x}'|$ 处, 有 $r \approx R - \boldsymbol{e}_R \cdot \boldsymbol{x}'$.

$$\begin{aligned}
\boldsymbol{A}(\boldsymbol{x}) &= \frac{\mu_0}{4\pi} \int \frac{\boldsymbol{J}(\boldsymbol{x}')\mathrm{e}^{\mathrm{i}k(R - \boldsymbol{e}_R \cdot \boldsymbol{x}')}}{R - \boldsymbol{e}_R \cdot \boldsymbol{x}'} \, \mathrm{d}^3 x' = \frac{\mu_0 \mathrm{e}^{\mathrm{i}kR}}{4\pi R} \int \boldsymbol{J}(\boldsymbol{x}')\mathrm{e}^{-\mathrm{i}k\boldsymbol{e}_R \cdot \boldsymbol{x}'} \, \mathrm{d}^3 x' \\
&= \frac{\mu_0 \mathrm{e}^{\mathrm{i}kR}}{4\pi R} \int \boldsymbol{J}(\boldsymbol{x}')(1 - \mathrm{i}k\boldsymbol{e}_R \cdot \boldsymbol{x}' + \cdots) \, \mathrm{d}^3 x' = \boldsymbol{A}^{(0)}(\boldsymbol{x}) + \boldsymbol{A}^{(1)}(\boldsymbol{x}) + \cdots \\
&= \frac{\mu_0 \mathrm{e}^{\mathrm{i}kR}}{4\pi R} \int \boldsymbol{J}(\boldsymbol{x}') \, \mathrm{d}^3 x' - \frac{\mathrm{i}k\mu_0 \mathrm{e}^{\mathrm{i}kR}}{4\pi R} \int \boldsymbol{J}(\boldsymbol{x}')(\boldsymbol{e}_R \cdot \boldsymbol{x}') \, \mathrm{d}^3 x' + \cdots
\end{aligned} \tag{5.128}$$

### 5.7.5 电偶极辐射

**Lem. 8**

若在边界处 $\boldsymbol{n} \cdot \boldsymbol{J}(\boldsymbol{x}) = 0$, 则有

$$\int \boldsymbol{J}(\boldsymbol{x}) \, \mathrm{d}^3 x = -\int \boldsymbol{\nabla} \cdot \boldsymbol{J}(\boldsymbol{x})\boldsymbol{x} \, \mathrm{d}^3 x \tag{5.129}$$

**Pf** 据 $\displaystyle\int \boldsymbol{\nabla} \cdot \boldsymbol{T} = \oint \boldsymbol{n} \cdot \boldsymbol{T} \, \mathrm{d}^2 x$, 有

$$\begin{aligned}
I &= \int \boldsymbol{\nabla} \cdot [\boldsymbol{J}(\boldsymbol{x})\boldsymbol{x}] \, \mathrm{d}^3 x = \int [\boldsymbol{\nabla} \cdot \boldsymbol{J}(\boldsymbol{x})]\boldsymbol{x} \, \mathrm{d}^3 x + \int \boldsymbol{J}(\boldsymbol{x}) \cdot \boldsymbol{\nabla}\boldsymbol{x} \, \mathrm{d}^3 x \\
&= \int \boldsymbol{\nabla} \cdot \boldsymbol{J}(\boldsymbol{x})\boldsymbol{x} \, \mathrm{d}^3 x + \int \boldsymbol{J}(\boldsymbol{x}) \, \mathrm{d}^3 x = \oint \underbrace{\boldsymbol{n} \cdot \boldsymbol{J}(\boldsymbol{x})}_{=0} \boldsymbol{x} \, \mathrm{d}^2 x = 0
\end{aligned}$$

**Def. 18　时谐电流体系的电偶极矩**

$$\boldsymbol{p}(t)=\boldsymbol{p}\mathrm{e}^{-\mathrm{i}\omega t}=\mathrm{e}^{-\mathrm{i}\omega t}\int\rho(\boldsymbol{x}')\boldsymbol{x}'\,\mathrm{d}^3x'=\mathrm{e}^{-\mathrm{i}\omega t}\frac{\mathrm{i}}{\omega}\int\boldsymbol{J}(\boldsymbol{x}')\,\mathrm{d}^3x' \tag{5.130}$$

**Rmk**

$$\dot{\boldsymbol{p}}=\frac{\mathrm{d}\boldsymbol{p}(t)}{\mathrm{d}t}=\mathrm{e}^{-\mathrm{i}\omega t}\int\boldsymbol{J}(\boldsymbol{x}')\,\mathrm{d}^3x' \tag{5.131}$$

**Pf**　$\displaystyle\int\rho(\boldsymbol{x}')\boldsymbol{x}'\,\mathrm{d}^3x'=\frac{1}{\mathrm{i}\omega}\int\boldsymbol{\nabla}'\cdot\boldsymbol{J}(\boldsymbol{x}')\boldsymbol{x}'\,\mathrm{d}^3x'=\frac{\mathrm{i}}{\omega}\int\boldsymbol{J}(\boldsymbol{x}')\,\mathrm{d}^3x'.$

**Prop. 32　电偶极子的辐射矢势**

$$\boldsymbol{A}^{(0)}(t,\boldsymbol{x})=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\dot{\boldsymbol{p}}(t) \tag{5.132}$$

**Rmk**　由 $\displaystyle\boldsymbol{\nabla}\frac{\mathrm{e}^{\mathrm{i}kR}}{R}=\left(-\frac{\boldsymbol{e}_R}{R^2}+\frac{\mathrm{i}k\boldsymbol{e}_R}{R}\right)\mathrm{e}^{\mathrm{i}kR}\approx\mathrm{i}k\boldsymbol{e}_R\frac{\mathrm{e}^{\mathrm{i}kR}}{R}$ 可得

$$\boldsymbol{\nabla}\to\mathrm{i}k\boldsymbol{e}_R \tag{5.133}$$

**Prop. 33　电偶极子的辐射场**

$$\boldsymbol{B}^{(0)}(t,\boldsymbol{x})=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\mathrm{i}k\boldsymbol{e}_R\times\dot{\boldsymbol{p}}(t)=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi cR}\ddot{\boldsymbol{p}}(t)\times\boldsymbol{e}_R \tag{5.134}$$

$$\boldsymbol{E}^{(0)}(t,\boldsymbol{x})=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}(\ddot{\boldsymbol{p}}\times\boldsymbol{e}_R)\times\boldsymbol{e}_R \tag{5.135}$$

**Lem. 9**

设球坐标的极轴为 $z$ 轴, 则有

$$\boldsymbol{e}_z\times\boldsymbol{e}_R=\boldsymbol{e}_\varphi\sin\theta \tag{5.136}$$

**Prop. 34　球坐标中的辐射场**

设电偶极矩沿极轴 $z$ 方向振荡 $\boldsymbol{p}=\boldsymbol{e}_zp$

$$\boldsymbol{B}^{(0)}(t,\boldsymbol{x})=-\frac{\mu_0\omega^2}{4\pi cR}\mathrm{e}^{\mathrm{i}(kR-\omega t)}p\sin\theta\,\boldsymbol{e}_\varphi=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi cR}\ddot{p}\sin\theta\,\boldsymbol{e}_\varphi \tag{5.137}$$

由 $\boldsymbol{e}_\varphi\times\boldsymbol{e}_R=\boldsymbol{e}_\theta$ 可得

$$\boldsymbol{E}^{(0)}(t,\boldsymbol{x})=-c\boldsymbol{e}_R\times\boldsymbol{B}^{(0)}(t,\boldsymbol{x})=-\frac{\mu_0\omega^2}{4\pi R}\mathrm{e}^{\mathrm{i}(kR-\omega t)}p\sin\theta\,\boldsymbol{e}_\theta=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\ddot{p}\sin\theta\,\boldsymbol{e}_\theta \tag{5.138}$$

**Prop. 35 辐射能流**

时谐电偶极子辐射场的 Poynting 矢量为

$$\begin{aligned} S(t,R,\theta) &= \frac{1}{\mu_0}\mathrm{Re}\left\{\boldsymbol{E}^{(0)}(t,R,\theta)\right\}\times\mathrm{Re}\left\{\boldsymbol{B}^{(0)}(t,R,\theta)\right\} \\ &= \frac{\mu_0\omega^4p^2}{16\pi^2cR^2}\sin^2\theta\cos^2(kR-\omega t)\boldsymbol{e}_R \end{aligned} \tag{5.139}$$

$$\langle\boldsymbol{S}\rangle(R,\theta) = \frac{\mu_0\omega^4p^2}{32\pi^2cR^2}\sin^2\theta\,\boldsymbol{e}_R \tag{5.140}$$

**Rmk** 辐射能流密度的角分布函数为

$$f(\theta)=\sin^2\theta \tag{5.141}$$

**Prop. 36 辐射功率**

$$P=\frac{\mu_0}{12\pi c}\omega^4p^2 \tag{5.142}$$

**Pf** $P=\oint\langle S\rangle\,\mathrm{d}^2x$, $I=\int_{\theta=0}^{\pi}\int_{\varphi=0}^{2\pi}\frac{\sin^2\theta}{R^2}R^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi=2\pi\int_0^{\pi}\sin^3\theta\,\mathrm{d}\theta=\frac{8\pi}{3}$.

**Cor. 10 时谐电偶极子远场区辐射场的特征**

1. 辐射电磁波近似为 TEM 波;
2. 场量振幅 $E,B\propto\dfrac{1}{R}$;
3. 在 $4\pi$ 的立体角内, 总辐射功率守恒, 与距离 $R$ 无关;
4. 能流密度与辐射功率 $S,P\propto\omega^4$.

**Eg. 1 中心馈电细短直天线的远场辐射**

设天线长度 $l\ll\lambda$, 电流为

$$I(t,z)=I_0\left(1-\frac{2}{l}|z|\right)\mathrm{e}^{-\omega t}\quad(|z|\leq l/2) \tag{5.143}$$

电偶极矩为

$$\boldsymbol{p}=\frac{\mathrm{i}}{\omega}\int_{-l/2}^{l/2}I(z)\,\mathrm{d}z\,\boldsymbol{e}_z=\frac{\mathrm{i}}{2\omega}I_0l\boldsymbol{e}_z \tag{5.144}$$

辐射功率为

$$P=\frac{\pi}{12}Z_0I_0^2\left(\frac{l}{\lambda}\right)^2 \tag{5.145}$$

定义等效电阻 $R_\mathrm{r}$, 满足 $P=\dfrac{1}{2}I_0^2R_\mathrm{r}$. 设 $\kappa=l/\lambda$, 则有

$$R_\mathrm{r}=\frac{\pi}{6}Z_0\left(\frac{l}{\lambda}\right)^2=\kappa^2 197\,\Omega \tag{5.146}$$

**Eg. 2 无辐射体系**

1. 稳恒源: $\dfrac{\partial\rho}{\partial t}=\dfrac{\partial \boldsymbol{J}}{\partial t}=0$, 电磁场不随时间变化, 因此无辐射.
2. 球对称径向运动: 球对称性使所有电磁多极矩均为零, 因此无辐射.

### 5.7.6 磁偶极辐射与电四极辐射

**Def. 19 时谐电流体系的磁偶极矩与电四极矩**

$$\boldsymbol{m}=\frac{1}{2}\int \boldsymbol{x}'\times\boldsymbol{J}(\boldsymbol{x}')\,\mathrm{d}^3x' \tag{5.147}$$

$$\mathcal{D}=\frac{1}{2}\int \rho(\boldsymbol{x}')\boldsymbol{x}'\boldsymbol{x}'\,\mathrm{d}^3x' \tag{5.148}$$

**Lem. 10**

$$\int \boldsymbol{J}\boldsymbol{x}+\boldsymbol{x}\boldsymbol{J}\,\mathrm{d}^3x=-\int(\boldsymbol{\nabla}\cdot\boldsymbol{J})\boldsymbol{x}\boldsymbol{x}\,\mathrm{d}^3x \tag{5.149}$$

**Pf** $\boldsymbol{\nabla}\cdot(\boldsymbol{J}\boldsymbol{x}\boldsymbol{x})=(\boldsymbol{\nabla}\cdot\boldsymbol{J})\boldsymbol{x}\boldsymbol{x}+\boldsymbol{J}\cdot\underbrace{\boldsymbol{\nabla}\boldsymbol{x}}_{=\mathcal{I}}\boldsymbol{x}+\boldsymbol{x}\boldsymbol{J}\cdot\underbrace{\boldsymbol{\nabla}\boldsymbol{x}}_{=\mathcal{I}}=(\boldsymbol{\nabla}\cdot\boldsymbol{J})\boldsymbol{x}\boldsymbol{x}+\boldsymbol{J}\boldsymbol{x}+\boldsymbol{x}\boldsymbol{J}$.

**Prop. 37 磁偶极辐射与电四极辐射的矢势**

$\boldsymbol{A}^{(1)}(\boldsymbol{x})$ 可利用二阶张量的对称化与反对称化分为两项

$$\boldsymbol{A}^{(1)}(\boldsymbol{x})=-\mathrm{i}k\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\boldsymbol{e}_R\cdot\int\boldsymbol{x}'\boldsymbol{J}\,\mathrm{d}^3x=-\mathrm{i}k\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\boldsymbol{e}_R\cdot\int\underbrace{\frac{1}{2}(\boldsymbol{x}'\boldsymbol{J}-\boldsymbol{J}\boldsymbol{x}')}_{\mathrm{AS}}+\underbrace{\frac{1}{2}(\boldsymbol{x}'\boldsymbol{J}+\boldsymbol{J}\boldsymbol{x}')}_{\mathrm{S}}\,\mathrm{d}^3x \tag{5.150}$$

反对称部分对应磁偶极辐射场的矢势

$$\boldsymbol{A}_\mathrm{m}^{(1)}(t,\boldsymbol{x})=\boldsymbol{A}_\mathrm{AS}^{(1)}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}=\mathrm{i}k\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\boldsymbol{e}_R\times\boldsymbol{m}(t) \tag{5.151}$$

对称部分对应电四极辐射场的矢势

$$\boldsymbol{A}_\mathrm{D}^{(1)}(t,\boldsymbol{x})=\boldsymbol{A}_\mathrm{S}^{(1)}(\boldsymbol{x})\mathrm{e}^{-\mathrm{i}\omega t}=\mathrm{i}k\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}[\mathrm{i}\omega\boldsymbol{e}_R\cdot\mathcal{D}(t)]=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi cR}[\boldsymbol{e}_R\cdot\ddot{\mathcal{D}}(t)] \tag{5.152}$$

于是可将矢势展开的第二极写为

$$\boldsymbol{A}^{(1)}(t,\boldsymbol{x})=\boldsymbol{A}_{\mathrm{m}}^{(1)}(t,\boldsymbol{x})+\boldsymbol{A}_{\mathrm{D}}^{(1)}(t,\boldsymbol{x})=\mathrm{i}k\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi R}\left(\boldsymbol{e}_R\times\boldsymbol{m}-\boldsymbol{e}_R\cdot\dot{\mathcal{D}}\right)\tag{5.153}$$

**Rmk** 交变电流情形有 $\boldsymbol{\nabla}\cdot\boldsymbol{J}=\mathrm{i}\omega\rho$, 因此电流体系可同时产生电多极辐射与磁多极辐射.

**Pf** $\dfrac{1}{2}\displaystyle\int(\boldsymbol{e}_R\cdot\boldsymbol{x}')\boldsymbol{J}-(\boldsymbol{e}_R\cdot\boldsymbol{J})\boldsymbol{x}'\,\mathrm{d}^3x'=-\boldsymbol{e}_R\times\left(\frac{1}{2}\int\boldsymbol{x}'\times\boldsymbol{J}\,\mathrm{d}^3x'\right)=-\boldsymbol{e}_R\times\boldsymbol{m}.$

$\dfrac{1}{2}\displaystyle\int\boldsymbol{J}\boldsymbol{x}'\boldsymbol{x}'\boldsymbol{J}\,\mathrm{d}^3x'=-\frac{1}{2}\int(\boldsymbol{\nabla}'\cdot\boldsymbol{J})\boldsymbol{x}'\boldsymbol{x}'\,\mathrm{d}^3x'=-\mathrm{i}\omega\frac{1}{2}\int\rho(\boldsymbol{x}')\boldsymbol{x}'\boldsymbol{x}'\,\mathrm{d}^3x'=-\mathrm{i}\omega\mathcal{D}.$

**Prop. 38 磁偶极辐射场的场量**

$$\boldsymbol{B}_{\mathrm{m}}^{(1)}(t,\boldsymbol{x})=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi c^2R}[\ddot{\boldsymbol{m}}(t)\times\boldsymbol{e}_R]\times\boldsymbol{e}_R=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi c^2R}\ddot{m}\sin\theta\,\boldsymbol{e}_\theta\tag{5.154}$$

$$\boldsymbol{E}_{\mathrm{m}}^{(1)}(t,\boldsymbol{x})=-\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi cR}\ddot{\boldsymbol{m}}(t)\times\boldsymbol{e}_R=-\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi cR}\ddot{m}\sin\theta\,\boldsymbol{e}_\varphi\tag{5.155}$$

**Rmk** 可由电偶极辐射场作代换得到磁偶极辐射场:

$$\begin{cases}\boldsymbol{p}\to\dfrac{\boldsymbol{m}}{c}\\ \boldsymbol{E}\to c\boldsymbol{B}\\ c\boldsymbol{B}\to-\boldsymbol{E}\end{cases}\tag{5.156}$$

**Prop. 39 磁偶极辐射场的能流密度与辐射功率**

$$\langle\boldsymbol{S}_{\mathrm{m}}\rangle=\frac{\mu_0\omega^4m^2}{32\pi^2c^3R^2}\sin^2\theta\,\boldsymbol{e}_R\tag{5.157}$$

$$P_{\mathrm{m}}=\frac{\mu_0\omega^4m^2}{12\pi c^3}\tag{5.158}$$

**Prop. 40 电四极辐射场的场量**

$$\boldsymbol{B}_{\mathrm{D}}^{(1)}(t,\boldsymbol{x})=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi c^2R}[\boldsymbol{e}_R\times\dddot{\mathcal{D}}(t)]\times\boldsymbol{e}_R\tag{5.159}$$

$$\boldsymbol{E}_{\mathrm{D}}^{(1)}(t,\boldsymbol{x})=\frac{\mu_0\mathrm{e}^{\mathrm{i}kR}}{4\pi cR}\{[\boldsymbol{e}_R\cdot\dddot{\mathcal{D}}(t)]\,\boldsymbol{\nabla}\times\boldsymbol{e}_R\}\,\boldsymbol{e}_R\tag{5.160}$$

**Prop. 41 电四极辐射场的能流密度**

$$\langle\boldsymbol{S}_{\mathrm{D}}\rangle=\frac{1}{4\pi\varepsilon_0}\frac{1}{8\pi c^5R^2}|[\boldsymbol{e}_R\cdot\dddot{\mathcal{D}}(t)]\times\boldsymbol{e}_R|^2\boldsymbol{e}_R\tag{5.161}$$

### 5.7.7 天线辐射

**Prop. 42 细长直天线的电流形式**

设天线沿 $z$ 轴, 则天线表面电流为 $\boldsymbol{J} = J_z \boldsymbol{e}_z$, 矢势 $\boldsymbol{A}(\boldsymbol{x}) = \dfrac{\mu_0}{4\pi}\displaystyle\int \frac{\boldsymbol{J}\mathrm{e}^{\mathrm{i}kr}}{r}\,\mathrm{d}^3x' = A_z\boldsymbol{e}_z$. Lorenz 规范化为

$$\frac{\partial A_z}{\partial z} + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = 0 \tag{5.162}$$

由天线表面电场 $E_z = 0$, 可得 $A_z$ 满足的波动方程

$$\frac{\partial^2 A_z}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2 A_z}{\partial t^2} = 0 \tag{5.163}$$

结合电流边界条件 $\boldsymbol{J}(\boldsymbol{x}')|_{\text{endpoints}} = 0$, 从而 $\boldsymbol{J}(\boldsymbol{x}')$ 有驻波形式.

**Prop. 43 对称细长直天线的辐射**

对称细长直天线的电流分布为

$$I(t,z) = I_0 \sin k(l/2 - |z|)\mathrm{e}^{-\mathrm{i}\omega t} \quad (|z| \leq l/2) \tag{5.164}$$

其矢势分量 $A_z$ 为

$$\begin{aligned}
A_z(t,\boldsymbol{x}) &= \frac{\mu_0}{4\pi}\int_{-l/2}^{l/2}\frac{I(t_{\text{rev}},z)}{r}\,\mathrm{d}z = \frac{\mu_0 I_0}{4\pi}\int_{-l/2}^{l/2}\frac{\sin k(l/2-|z|)}{r}\mathrm{e}^{-\mathrm{i}\omega(t-r/c)} \\
&\approx \frac{\mu_0 I_0}{4\pi R}\mathrm{e}^{\mathrm{i}(kR-\omega t)}\int_{-l/2}^{l/2}\sin\left(\frac{kl}{2} - l|z|\right)\mathrm{e}^{-\mathrm{i}kz\cos\theta}\,\mathrm{d}z \\
&= \frac{\mu_0 I_0}{2\pi kR}\mathrm{e}^{\mathrm{i}(kR-\omega t)}\frac{1}{\sin^2\theta}\left(\cos\frac{kl\cos\theta}{2} - \cos\frac{kl}{2}\right)
\end{aligned} \tag{5.165}$$

其辐射电磁场为

$$\boldsymbol{B}(t,\boldsymbol{x}) = -\mathrm{i}\frac{\mu_0 I_0}{2\pi R}\mathrm{e}^{\mathrm{i}(kR-\omega t)}\frac{1}{\sin\theta}\left(\cos\frac{kl\cos\theta}{2} - \cos\frac{kl}{2}\right)\boldsymbol{e}_\varphi \tag{5.166}$$

$$\boldsymbol{E}(t,\boldsymbol{x}) = -\mathrm{i}\frac{\mu_0 I_0 c}{2\pi R}\mathrm{e}^{\mathrm{i}(kR-\omega t)}\frac{1}{\sin\theta}\left(\cos\frac{kl\cos\theta}{2} - \cos\frac{kl}{2}\right)\boldsymbol{e}_\theta \tag{5.167}$$

辐射能流密度为

$$\langle \boldsymbol{S}\rangle = \frac{\mu_0 I_0 c}{8\pi^2 R^2}\underbrace{\frac{1}{\sin^2\theta}\left(\cos\frac{kl\cos\theta}{2} - \cos\frac{kl}{2}\right)^2}_{=f(\theta)}\boldsymbol{e}_R \tag{5.168}$$

**Eg. 3 半波天线**

取对称细长直天线长度为 $l=\nu\dfrac{\lambda}{2}$, 则称之为半波天线. 对于 $\nu=1$ 的半波天线:

$$I(t,z)=I_0\cos(kz)\mathrm{e}^{-\mathrm{i}\omega t}\quad(|z|\le\lambda/4)\tag{5.169}$$

$$\boldsymbol{A}(t,\boldsymbol{x})=\frac{\mu_0 I_0}{2\pi kR}\mathrm{e}^{\mathrm{i}(kR-\omega t)}\frac{\cos\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin^2\theta}\boldsymbol{e}_z\tag{5.170}$$

$$\boldsymbol{B}(t,\boldsymbol{x})=-\mathrm{i}\frac{\mu_0 I_0}{2\pi R}\mathrm{e}^{\mathrm{i}(kR-\omega t)}\frac{\cos\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin\theta}\boldsymbol{e}_\varphi\tag{5.171}$$

$$\boldsymbol{E}(t,\boldsymbol{x})=-\mathrm{i}\frac{\mu_0 I_0 c}{2\pi R}\mathrm{e}^{\mathrm{i}(kR-\omega t)}\frac{\cos\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin\theta}\boldsymbol{e}_\theta\tag{5.172}$$

$$\langle\boldsymbol{S}\rangle=\frac{\mu_0 I_0^2 c}{8\pi^2R^2}\frac{\cos^2\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin^2 R}\tag{5.173}$$

$$f(\theta)=\frac{\cos^2\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin^2\theta}\tag{5.174}$$

$$P=2.44\frac{\mu_0 I_0^2 c}{8\pi}\tag{5.175}$$

$$R_{\mathrm{r}}=2.44\frac{\mu_0 c}{4\pi}=73.2\,\Omega\tag{5.176}$$

**Prop. 44 天线阵**

# 6　运动带电粒子与电磁场的相互作用

## 6.1　带电粒子的辐射

### 6.1.1　运动带电粒子的势

**Def. 1　Liénard-Wiechert 势**

$$\begin{cases} \boldsymbol{A} = \dfrac{q\boldsymbol{v}}{4\pi\varepsilon_0 c^2\left(r - \dfrac{\boldsymbol{v}}{c}\cdot\boldsymbol{r}\right)} \\ \phi = \dfrac{q}{4\pi\varepsilon_0\left(r - \dfrac{\boldsymbol{v}}{c}\cdot\boldsymbol{r}\right)} \end{cases} \tag{6.1}$$